

RESEARCH REPORT

New Trends in Science and Technology Innovation Policy from the Perspective of Science and Technology Diplomacy (2025)

調査報告書

科学技術外交からみた科学技術
イノベーション政策の新潮流 (2025年)

Introduction

This report provides an overview of science and technology innovation (STI) policy from the perspective of science and technology diplomacy, which is rapidly gaining importance and interest in the midst of changing global circumstances, interactions among nations, international frameworks, and trends in international organizations and forums that operate beyond national boundaries.

Ever since the emergence of modern science in the 17th century, various initiatives have been taken that both promote science and are motivated by diplomatic objectives. These efforts were conceptualized in a report titled “New Frontiers in Science Diplomacy”¹ published in January 2010, based on discussions at an international workshop held by The Royal Society and the American Association for the Advancement of Science (AAAS). This report described the three pillars of science diplomacy in terms of science in diplomacy, diplomacy for science, and science for diplomacy. Since then, various efforts have been made in science and technology diplomacy through science and technology innovation policies and foreign policies. Most recently, science and technology diplomacy affected by changes in geopolitical situations and the accelerating rate of progress in science and technology are having a dramatic effect on society and politics. Emerging technologies such as AI, quantum technology, semiconductors, and biotechnology are having a large impact as the results of basic research are being implemented in society in a very short time. Science and technology are no longer limited to the traditional objectives of solving social and economic problems, but are now deeply related to politics and national power. Due to the need for science and technology to address not only economic competitiveness but also global issues such as sustainability, resilience, quality of life, and security, STI policies continue to change while overlapping with industrial, security, and foreign policies, and STI-related organizations around the world are trying to establish science, technology, and innovation ecosystems through various initiatives and collaborations.

In light of these rapid changes in the global situation, the importance of science and technology diplomacy is increasing, and moves to explore new concepts, roles, and initiatives to adapt science and technology diplomacy to the changes that are taking place are gaining momentum. For example, the European Commission is promoting the development of an EU framework for science diplomacy. In January 2024, The Royal Society and AAAS announced in “Science Diplomacy – 15 Years On”² that they would be making a year-long review of the three pillars of science and technology diplomacy proposed in 2010 and how they have been applied on various occasions since then, and in February 2025, a new report, “Science Diplomacy in an

1 AAAS, “New Frontiers in Science Diplomacy,” https://www.aaas.org/sites/default/files/New_Frontiers.pdf

2 AAAS, “Science Diplomacy — 15 Years On,” <https://www.sciencediplomacy.org/editorial/2024/science-diplomacy-15-years>

Era of Disruption”³ was issued. This report proposed a simplified two-dimensional framework: Science impacting diplomacy—how science engages with diplomatic objectives, and diplomacy impacting science—how diplomacy engages with and influences scientific activities. By simplifying in this way, the aim is to create a more flexible framework that is adaptable to current and future challenges. The three key points for practicing science diplomacy are: (1) science diplomacy is an important tool for promoting international relations, (2) science diplomacy is increasingly being used by non-state actors such as private companies, and (3) science diplomacy is required in times of turmoil. Furthermore, interest in science and technology diplomacy is growing in the Global South, including India and ASEAN countries, and in March 2025, UNESCO is planning a ministerial-level meeting on science and technology diplomacy.

This report updates and expands parts of the “Global Trends in Science, Technology and Innovation Policy (2024)” and adds new trends. We hope that this report will help you recognize the growing global importance of science and technology in national strategies and policies from the perspective of science and technology diplomacy, and help you consider what measures Japan should adopt. Since the term “Science and Technology Diplomacy” is commonly used in Japan, while the term “Science Diplomacy” is used in Europe and the United States, this report will treat “Science and Technology Diplomacy” and “Science Diplomacy” in Europe and the United States as one and the same.

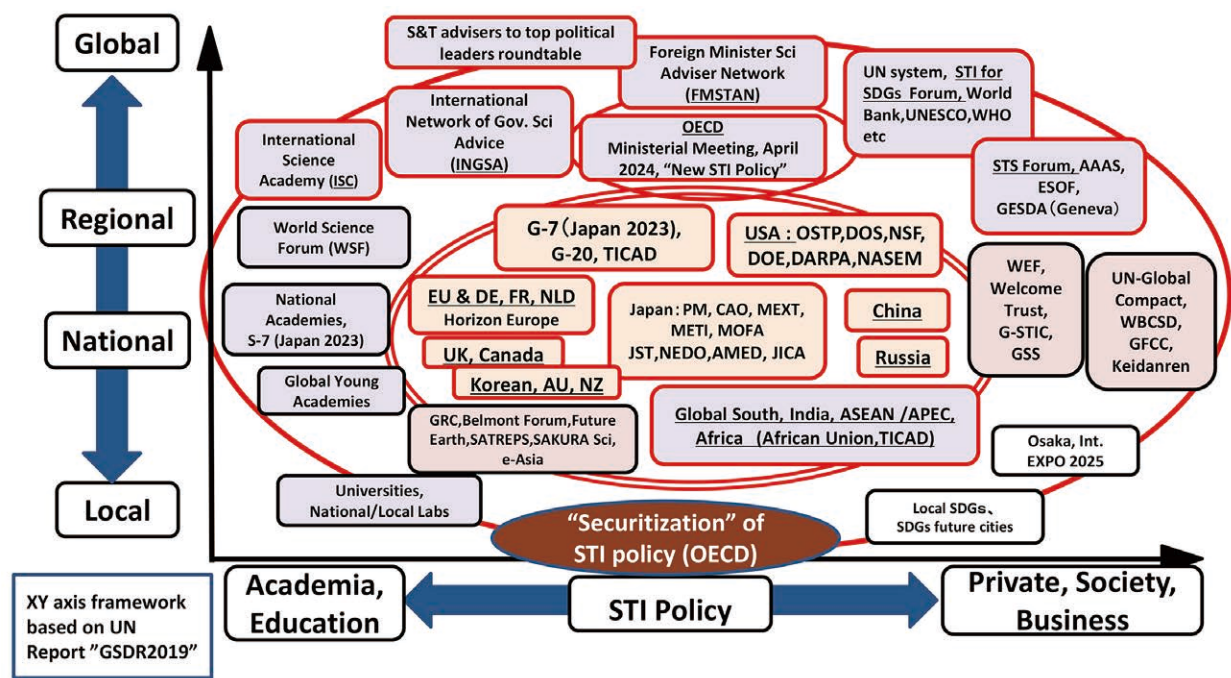


Figure 1 Global correlation of STI-related organizations
Prepared by CRDS based on various sources

3 AAAS, “Science diplomacy in an era of disruption,”
<https://www.aaas.org/news/science-diplomacy-era-disruption>

* The annotations in this document are based on confirmed access as of February 2025.

* Japan Science and Technology Agency (JST) is referred to as JST

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1 | Science & Technology Diplomacy

1.1 How the Concept of Science and Technology Diplomacy was Formed

Since the advent of modern science in the 17th century, various initiatives have been pursued, both to promote science and for diplomatic motives. These initiatives include, for example, international research cooperation, the establishment of international rules for the use of nuclear energy, space, and ocean development after World War II, and the operation of large international research facilities, such as the European Organization for Nuclear Research (CERN¹, Geneva, Switzerland) established in 1954, the International Institute for Applied Systems Analysis (IIASA², Vienna, Austria) established in 1964, ITER Organization (ITER, Cadarache, France³), the Human Frontier Science Program (HFSP⁴) proposed by Japan in 1987 and launched in 1990, the International Space Station (ISS⁵), construction of which began in 1998 and was completed in 2011, the Intergovernmental Panel on Climate Change (IPCC⁶), established in 1988, and the Synchrotron-light for Experimental Science and Applications in the Middle East (SESAME, Alan, Jordan)⁷, which was officially opened in 2017. In addition, since 1995, member countries of the United Nations Framework Convention on Climate Change (UNFCCC), the international treaty on climate change, have annually convened their own supreme decision-making body, known as the Conference of the Parties (COP). The IPCC provides scientific findings to the COP, which in turn uses these findings as the basis for national policy discussions and consensus building on specific policies. The IPCC's capabilities and track record in bridging scientific knowledge and international political decision-making provides a model for building mechanisms for scientific and political relationships on various other global issues, such as biodiversity.

A major factor in the conceptualization of these efforts as “science and technology diplomacy” and their revitalization under science and technology innovation policy and foreign policy was a report published in January 2010, based on discussions at an international workshop held by The Royal Society and the American Association for the Advancement of Science (AAAS). The report is titled “New Frontiers in Science Diplomacy”⁸ and states, “Science diplomacy is not new,

- 1 CERN, “CERN” <https://home.cern/>
- 2 IIASA, “International Institute for Applied Systems Analysis,” <https://iiasa.ac.at/>
- 3 QST, “ITER Progress” https://www.fusion.qst.go.jp/ITER/iter/page1_1.html
- 4 International Human Frontier Science Program Organization (HFSP), “HFSP,” <https://www.hfsp.org/>
- 5 JAXA, “ISS” <https://humans-in-space.jaxa.jp/iss/>
- 6 Japan Meteorological Agency, “Intergovernmental Panel on Climate Change” <https://www.data.jma.go.jp/cpdinfo/ipcc/index.html>
- 7 SESAME, “SESAME” <https://www.sesame.org.jo/>
- 8 AAAS, “New Frontiers in Science Diplomacy,” https://www.aaas.org/sites/default/files/New_Frontiers.pdf

but it is more important than ever. Many of the issues that will define the 21st century, from climate change and food security to poverty reduction and nuclear disarmament, have a scientific dimension. No country can solve these problems on its own. The tools, techniques, and tactics of foreign policy must adapt to an increasingly scientifically and technologically complex world.” The report went further to propose three elements of science diplomacy: science in diplomacy, diplomacy for science, and science for diplomacy. Based on this, Europe, the U.S., and Japan have made contributions and established networks in various international frameworks, including the appointment of science advisors to heads of state, foreign ministers, and other ministers, and, in Japan, the creation of science-related positions such as science diplomacy envoys and technology ambassadors within the Ministry of Foreign Affairs (Figure 2).

Figure 2 Major events related to science and technology diplomacy

Japan Events	World Events
1960s-80sThe Dawn of Scientific Advice	
Treaty of Mutual Cooperation and Security between the United States and Japan(1960) Pollution problems became more serious with rapid economic growth: Basic Act on Pollution Control (1967), Air Pollution Control Act (1968), and establishment of the Environmental Agency (1971)	UK appoints Government Chief Scientific Adviser (1964) The United States establishes the Office of Science and Technology Policy (OSTP) in the Office of the President, evolving from the Office of Science and Technology organized in 1961.(1976) Director of OSTP serves as Science Advisor to the President (scope of authority varies by administration) <i>The Silent Spring</i> (1962) by R. Carson and “The Limits to Growth (1972)” Club of Rome Report are published as environmental issues became more apparent. Growing international demand for a response to climate change Intergovernmental Panel on Climate Change IPCC established (1988), UN Conference on Sustainable Development (Rio+20) (1992) The Rio Declaration on Environment and Development is adopted, and the United Nations Framework Convention on Climate Change and the Convention on Biological Diversity is signed.
1990s Society’s Expectations and Distrust of Science and Technology	
Enactment of the Science and Technology Basic Law (1995) First Science and Technology Basic Plan (1996) Growing distrust of science and technology: Monju sodium leak accident (1995), sarin gas attack on the subway (1995), Tokai-mura JOC criticality accident (1999), etc.	Human BSE infection in the U.K. becomes a social problem (1996) Kyoto Protocol adopted at COP3 (held annually since 1997) World Science Forum (WSF) established with Budapest Declaration (1999) advocating “Science for Society”, meetings held every other year since 2003
2000s The Dawn of Science and Technology Diplomacy	
Establishment of the Council for Science and Technology Policy (2001), Incorporation of national universities (2004), Science and Technology in Society Forum (STS Forum) launched (2004), Science and Technology Research Partnership for Sustainable Development (SATREPS) launched (2008)	The United States establishes the first Science and Technology Advisor to the Secretary of State (1999) The United Kingdom establishes the first Chief Scientific Adviser to the Foreign and Commonwealth Office (2009)

Japan Events	World Events
2010s Promoting Science and Technology Diplomacy	
Great East Japan Earthquake (2011) The term “Science and Technology Diplomacy” appeared in the 4th Science and Technology Basic Plan (2012) and the Ministry of Foreign Affairs “Report of the Advisory Panel on Science and Technology Diplomacy” (2014) Appointment of Mr. Teruo Kishi as the first Science and Technology Advisor to the Minister of Foreign Affairs (2015). New diplomatic contributions, starting with the 1st Symposium on Science and Technology Diplomacy in Tokyo (2016), the G7 Ise-Shima Summit (2016), TICAD VI in Kenya (2016), and TICAD7 in Yokohama (2019), UN Multi-Stakeholder Forum on Science, Technology and Innovation for the Sustainable Development Goals (held annually since 2016), the G20 Osaka Summit agreement to create STI for SDGs roadmap (2019) 2nd Symposium on Science and Technology Diplomacy in Tokyo (2019)	The Royal Society and the American Association for the Advancement of Science (AAAS) advocate three types of science and technology diplomacy (2010) Establishment of the International Network for Governmental Science Advice (INGSA) (2014) UN Sustainable Development Goals (SDGs) adopted (2015) Foreign Ministries Science & Technology Advice Network (FMSTAN) established (2016) U.S. Trump administration comes to power (2017) → U.S.-China high-tech friction FMSTAN founding members advocate three actions for science and technology diplomacy (2018) International Council for Science (ICSU) and International Social Science Council (ISCC) merge to form International Science Council (ISC) (2018) 9th World Science Forum (WSF) in Budapest (2019) ⇒ Recognizing the importance of the Budapest Declaration of 20 years ago, advocates science that contributes to well-being, etc.
2020s Science and Technology Diplomacy in an Era of Geopolitical Tension	
Yoichiro Matsumoto appointed as the second Science and Technology Advisor to the Minister of Foreign Affairs (2020) TICAD 8 in Tunisia (2022) Mr. Kazuhito Hashimoto appointed as the first Science and Technology Advisor to the Cabinet of Japan (2022). He chaired the following international meetings: Science Advisers’ Roundtable 2022 in Kyoto, G7 Science Advisers’ Roundtable (2023 in the U.S.), Science Advisers’ Roundtable 2023 (in Kyoto) Japan assumes the G7 chairmanship (2023-2024), Science and Technology Ministers agree on the importance of international brain circulation and the promotion of trustworthy scientific research through research integrity and security initiatives. ASEAN-Japan Special Leaders’ Meeting (December 2023 in Tokyo), 3rd Science and Technology Diplomacy Symposium (2024 in Tokyo) Commencement of the 7th Science and Technology Innovation Strategy (FY2026-FY2030)	Outbreak of the coronavirus pandemic (2020-) Birth of the Biden administration (2021) → Return of the U.S. to science, technology, and international cooperation Arati Prabhakar appointed OSTP Director and Science Advisor to the President (2022) Angela MacLean appointed Chief Scientific Adviser to the UK Government (2023) Russian invasion of Ukraine (2022-) OECD Committee on Science and Technology Policy (CSTP) publishes its core report “STI Outlook 2023” (2023) OECD Ministerial Council adopts “Indo-Pacific Framework” (2023) Israel-Hamas conflict (2023-) First EU Science Diplomacy Conference (2023) OECD Science and Technology Policy Committee Ministerial Meeting (2024) Second Trump Administration (2025)

Prepared by CRDS based on various sources

1.2 Examining New Concepts of Science and Technology Diplomacy

In 2017, Teruo Kishi, then Science and Technology Advisor to Japan’s Ministry of Foreign Affairs, and four science advisors from the United States, the United Kingdom, and New Zealand jointly proposed a new conceptual framework for practice to replace the traditional taxonomy: 1)

actions directly related to pursuing national interests, 2) actions that address cross-border interests, and 3) actions that respond to global demands and challenges. In addition, they recommended that since science and technology diplomacy is an important element of diplomacy for all countries, regardless of their size or stage of development, governments should promote collaboration among related institutions, including diplomacy, trade, development cooperation, and security, and position science and technology as a major instrument⁹.

In light of the rapidly changing global situation, the importance of science and technology diplomacy is growing, and there is an active movement to explore new concepts, roles, and initiatives for science and technology diplomacy adapted to the ongoing changes (Figure 3). The European Commission is promoting the development of a science diplomacy framework for the EU. The Royal Society and AAAS published an editorial titled “Science Diplomacy – 15 years on” in January 2024, announcing that the three elements of science and technology diplomacy proposed in 2010 will be reviewed on various occasions over the next year. A special issue of *Science Diplomacy* was published in October 2024, and a new report, “Science Diplomacy in an Era of Disruption,” was issued in February 2025. The report proposed a simplified framework with two dimensions: Science impacting diplomacy: How science relates to diplomatic objectives, and Diplomacy impacting science: How diplomacy engages and influences scientific activities. The report laid out three important points for the practice of science diplomacy that aims to simplify and create a more flexible framework that is adaptable to current and future challenges: (1) Science diplomacy is an important tool to promote international relations, (2) Science diplomacy is increasingly being used by non-state actors, and (3) Science diplomacy is required in times of turmoil. (See Column 1 for a summary of the report.)

In addition, the Global South, including India and ASEAN countries, is also increasingly interested in science and technology diplomacy, and held a ministerial-level meeting on science and technology diplomacy at UNESCO in March 2025 (see Chapter 2, 2.5 for more on science and technology diplomacy in the Global South).

9 Gluckman, Peter D., Vaughan C. Turekian, Robin W. Grimes, and Teruo Kishi. “Science Diplomacy: A Pragmatic Perspective from the Inside.” *Science & Diplomacy* 6.4 (2017): 1–13.
https://www.sciencediplomacy.org/sites/default/files/pragmatic_perspective_science_advice_dec2017_1.pdf

Figure 3 Discussion forums related to science and technology diplomacy

Time of year	Contents
May 2023	G7 Leaders' Summit (in Hiroshima): Adoption of the Leaders' Communiqué on "Promoting International Mobility and Circulation of Human Resources" in science and technology.
September 2023	G20 Leaders' Summit (in India) "Creating a Global Science Advisory Mechanism with Science Diplomacy as a Tool", the outcome document of the G20 Chief Science Advisors' Roundtable adopted as an annex.
October 2023	STS Forum (in Kyoto) Then-Prime Minister Kishida mentions in his opening remarks that "Science diplomacy is indispensable." Science Advisers' Roundtable held.
December 2023	1st European Science Diplomacy Conference (in Madrid, Spain)
January 2024	In Science & Diplomacy, AAAS and The Royal Society announce a review of the science diplomacy framework for 2025 in the article "Science Diplomacy - 15 Years On".
February 2024	AAAS Annual Meeting (in Denver, USA). AAAS, EU, and others hold sessions on science and technology diplomacy
March 2024	The 3rd Symposium on Science and Technology Diplomacy (in Tokyo) organized by Ministry of Foreign Affairs of Japan and JST
April 2024	OECD Committee on Science and Technology Policy (CSTP) Ministerial Level Meeting Japan ASEAN High-Level Roundtable on Science, Technology and Innovation (side event)
May 2024	International Network for Governmental Science Advice (INGSA) meeting and Foreign Ministries Science & Technology Advice Network (FMSTAN) meeting (in Kigali, Rwanda)
May 2024	9th UN Multi-Stakeholder Forum on Science, Technology and Innovation for the Sustainable Development Goals (in NY, USA)
May 2024	GRC (Global Research Council) Annual Meeting (in Interlaken, Switzerland)
August 2024	Science Diplomacy Workshop for Early to Mid-Career Researchers in ASEAN Plus Three (Co-organized by AAAS, TWAS, and TSRI, in Thailand)
October 2024	STS Forum (in Kyoto) In a video message, Prime Minister Ishiba advocates accelerating international human resource exchange (international brain circulation) while ensuring research security amid the growing importance of international science and technology cooperation and science and technology diplomacy.
October 2024	Geneva Science and Diplomacy Anticipator (GESDA) Annual Meeting (in Geneva, Switzerland)
November 2024	World Science Forum 2024 (in Budapest, Hungary) Asia-Europe Foundation (ASEF) holds an event to report on the "Asia-Europe Science and Technology Diplomacy Report" (side event)
January 2025	2nd OECD Open Seminar in Tokyo, hosted by JST
February 2025	AAAS Annual Meeting (in Boston, USA), AAAS and The Royal Society Announce New Science and Technology Diplomacy Framework

Prepared by CRDS based on various sources

Column 1

A New Framework for Science and Technology Diplomacy: Science Diplomacy in an Era of Disruption (Royal Society/AAAS) Overview

** The report has been summarized by CRDS. Underlining added by CRDS.*

We are living in times of turmoil. The geopolitical landscape is becoming increasingly confrontational, power is more widely dispersed, and relations among major powers are more competitive. The global scientific and technological landscape is changing rapidly, and with it, science and technology are becoming even more important to national governments for economic growth and international competitiveness. Intertwined with this turmoil are environmental issues such as climate change and loss of biodiversity, which continue to pose significant threats to life on Earth. In these times of change and uncertainty, we need a framework for the practice of science diplomacy that accurately recognizes the real world, can be used by both state and non-state actors, and enables practitioners from diplomats to industrial leaders to address challenges and seize new opportunities in a confusing world.

In 2010, The Royal Society and the American Association for the Advancement of Science (AAAS) published a report titled “New Frontiers in Science Diplomacy”, which proposed the following three aspects as a framework for science diplomacy. Science in Diplomacy, Diplomacy for Science and Science for Diplomacy. This report is still widely cited and discussed today.

However, as the world becomes more fragmented and complex, effective science diplomacy must be increased to ensure that the conceptual framework is conducive to practical application. A science diplomacy framework should focus on the interaction between science and diplomacy. To this end, we propose a simpler framework that organizes the relationship between science and diplomacy into the following two aspects.

Two Dimensions of the Proposed Science Diplomacy Framework

Science impacting diplomacy: How science relates to diplomatic

objectives.

Diplomacy impacting science: How diplomacy engages and influences scientific activities.

These two dimensions do not imply that there are only two ways to engage science and diplomacy, but rather provide a framework that encompasses a variety of ways in which science engages with diplomacy and diplomacy engages with science.

By simplifying the framework, AAAS and The Royal Society aim to create a more flexible framework that is adaptable to current and future challenges. In considering the new framework, three key points regarding the practice of science diplomacy were also identified.

1. **Science diplomacy is an important tool to promote international relations:** Scientists seek to gain and apply new knowledge about the world. On the other hand, the practice of diplomacy always involves the pursuit of the interests of nations and organizations. When the interests of both parties are aligned, science and diplomacy can achieve great results, as many practical examples, such as climate change assessments and the construction and operation of large-scale infrastructure, demonstrate. In the 2010 report, the discussion revolves around these positive interactions, which may create the perception that “science diplomacy is always positive and a tool to be used for the global public good.” In practice, however, this is not always the case. Science diplomacy is a tool to achieve the diplomatic objectives of a nation or organization, and whether those objectives are perceived positively depends on one’s point of view.
2. **Science diplomacy is increasingly used by non-state actors:** Science diplomacy for national and international purposes has traditionally been implemented by diplomats representing national governments. In recent years, however, non-state actors, particularly global technology companies, have been using science diplomacy to develop their own “diplomacy”. This trend indicates that the interests of corporations may not necessarily coincide with those of nations.
3. **Times of turmoil call for scientific diplomacy:** Modern society is being profoundly transformed by innovations in science, engineering, and technology, including the rapid development of AI. As regulatory and governance frameworks fail to keep pace, the giant corporations that develop and supply these technologies are pursuing their own diplomatic activities. The open system of international scientific

cooperation is being used in some cases to strengthen the military capabilities of certain countries, raising concerns about the security of research. And, underdeveloped areas once considered the focus of scientists' research, such as the deep sea, the polar regions, the moon, and outer space, are now becoming the subject of political competition. In addition, measures to address global challenges such as climate change, loss of biodiversity, and global poverty remain inadequate. With more than 8 billion people living on the planet today, the greatest threats facing present and future generations remain unresolved. At these critical moments, science diplomacy will play an increasingly important role for both state and non-state actors.

Summary of Discussions

In preparing this report, input was received from a wide range of stakeholders through roundtables, conferences, and international events. The main points made are as follows. Science and diplomacy are increasingly closely linked. Science is becoming more important in foreign policy, and diplomacy increasingly needs to work with science. The interests of science and diplomacy do not necessarily coincide. Conflicts can arise because scientists pursue knowledge and diplomats pursue national interests. Scientific values that have been considered universal are being reevaluated. This has implications for the nature of international scientific cooperation and the practice of science diplomacy. Awareness of national security risks is required in scientific cooperation. The roles and responsibilities of practitioners of science diplomacy need to be clarified. Trust in science and scientific advice is challenged. Non-state actors play an important role in science diplomacy. More comprehensive international scientific cooperation is required.

2 | Country/Region Initiatives and Noteworthy Trends

2.1 Japan

In April 2007, at the then Council for Science and Technology Policy meeting, expert members submitted a proposal titled “Toward Strengthening Science and Technology Diplomacy”¹⁰, which stated, “It is important to enhance Japan’s soft power by making full use of its science and technology capabilities and actively and continuously addressing global challenges to realize a sustainable society, and to link research and technology cooperation with diplomacy.” In May 2008, a report with the same title¹¹ was compiled, which led to the establishment of the Science and Technology Research Partnership for Sustainable Development (SATREPS) to promote international science and technology cooperation using ODA. Through SATREPS, JST and the Japan Agency for Medical Research and Development (AMED), in collaboration with the Japan International Cooperation Agency (JICA), continue to promote international joint research and human resource development between Japan and developing countries to solve global-scale issues. In addition, JST and AMED have been supporting various types of international research cooperation in research funding programs related to science and technology diplomacy, such as the Strategic International Collaborative Research Program (SICORP, launched in 2009)¹² and the e-ASIA Collaborative Research Program (launched in 2012). In addition, most recently, ASPIRE (Adopting Sustainable Partnerships for Innovative Research Ecosystem)¹³ was launched in 2023, and the Japan-ASEAN Science, Technology and Innovation Cooperation Project (Networked Exchange, United Strength for Stronger Partnerships between Japan and ASEAN (NEXUS))¹⁴ was launched in 2024. ASPIRE aims to maintain and improve Japan’s science and technology capabilities by connecting top researchers from Japan and other advanced science and technology countries and regions through international joint research in seven policy-important science and technology fields (AI and information, biotechnology, energy, materials, quantum, semiconductors, and communications). NEXUS also aims to take advantage of the 50th anniversary of the ASEAN-Japan Friendship Cooperation to promote flexible and multilayered science and technology cooperation that leverages the strengths of both sides, to further strengthen mutual sustainable research cooperation, and to grow as a partner in co-creating innovation.

10 Cabinet Office, “Toward Strengthening Science and Technology Diplomacy.” <https://www8.cao.go.jp/cstp/siryo/haihu66/siryo1.pdf>

11 Cabinet Office, “Toward Strengthening Science and Technology Diplomacy.” <https://www8.cao.go.jp/cstp/output/080519iken-5.pdf>

12 JST, “Outline of SICORP” <https://www.jst.go.jp/inter/program/structure/general.html>

13 JST, “Outline of ASPIRE” <https://www.jst.go.jp/aspire/about.html>

14 JST, “Outline of NEXUS” <https://www.jst.go.jp/aspire/nexus/about/index.html>

At the policy level, the Fourth Science and Technology Basic Plan¹⁵, which was launched in 2012, includes the following four items under “New Development of Science and Technology Diplomacy”: (1) Development of international activities that leverage Japan’s strengths, (2) Promotion of international activities related to advanced science and technology, (3) Promotion of coordination and cooperation with developing countries on global-scale issues, and (4) Strengthening of the infrastructure for developing international activities in science and technology. The 6th Science, Technology and Innovation Basic Plan (FY2021-FY2025) calls for the promotion of strategic science and technology diplomacy, including “strategic cooperation and collaboration in key cutting-edge fields,” “strengthening the infrastructure supporting the strategic development of science and technology diplomacy,” and “building an international research network in which Japan plays a central role.” As of February 2025, the 7th Science, Technology and Innovation Basic Plan (FY2026 - FY2030) is under consideration. In formulating the 7th Science, Technology and Innovation Basic Plan, it is necessary to keep in mind that the surrounding environment is very different from that of the 6th Basic Plan formulated five years ago, including changes in the global situation and rapid development of emerging key technologies, and there is an increasing need to consider the plan in light of international trends.

One of the initiatives of the Ministry of Foreign Affairs of Japan is the establishment of the position of Science and Technology Advisor to the Minister of Foreign Affairs. In September 2015, then Foreign Minister Fumio Kishida appointed Teruo Kishi (Professor Emeritus, The University of Tokyo) as the first Science and Technology Advisor to the Minister of Foreign Affairs, based on the recommendation of the Advisory Panel of Experts on Science and Technology Diplomacy (chaired by Takashi Shiraishi, then President of National Graduate Institute for Policy Studies). Mr. Kishi served as Science and Technology Advisor to the Minister of Foreign Affairs for approximately four and a half years until March 2020, and Mr. Yoichiro Matsumoto (Professor Emeritus, University of Tokyo) was appointed as the second Science and Technology Advisor to the Minister of Foreign Affairs in April 2020. In the meantime, Mitsunobu Kano (Professor, Okayama University) served as Deputy Science and Technology Advisor to the Minister of Foreign Affairs from 2019-2022, and Motoko Kotani (Professor, Tohoku University) was appointed as his successor in 2022. Currently, Yoichiro Matsumoto and Motoko Kotani are engaged in a variety of important activities, including “the establishment of Science and Technology Fellows at diplomatic missions abroad”¹⁶ and “proposals on Science and Technology Diplomacy and ODA”¹⁷. In March 2023, the 3rd Science and Technology Diplomacy Symposium was held in Tokyo to discuss science and technology diplomacy in an era of change with

15 Cabinet Office, “Basic Plan for Science and Technology”
<https://www8.cao.go.jp/cstp/kihonkeikaku/4honbun.pdf>

16 Ministry of Foreign Affairs of Japan, “Establishment of Science and Technology Fellows in Overseas Establishments.” https://www.mofa.go.jp/mofaj/dns/isc/page22_004113.html

17 Ministry of Foreign Affairs of Japan, “Courtesy call on Foreign Minister Kamikawa by Yoichiro Matsumoto, Advisor to the Minister of Foreign Affairs on Science and Technology, etc.”
https://www.mofa.go.jp/mofaj/press/release/pressit_000001_00681.html

speakers from the US, Thailand, and the EU.

Column 2

Activities of Japan's First Science and Technology Advisor to the Minister of Foreign Affairs

Mr. Teruo Kishi, the first Science and Technology Advisor to the Minister of Foreign Affairs of Japan, was active for four and a half years. A summary of his activities follows.

Providing Scientific Advice to Diplomacy (Science in Diplomacy)

- Contribution to the Outcome Document of the G7 Ise-Shima Summit
- Two recommendations and concrete results for TICADVI and TICAD7
- Two recommendations and concrete results for promoting STI for SDGs
- Recommendations on the use of scientific knowledge in the Arctic
- Three papers co-authored with science and technology advisors from other countries
- Examination of the role of science and technology advisors through the holding of two symposiums on science and technology diplomacy

Dissemination of Japan's science and technology innovation trends both domestically and internationally (Diplomacy for Science)

- Events held by the Cabinet Office and the Ministry of Foreign Affairs (diplomatic missions abroad) to promote Japanese science and technology in other countries (SIP projects, etc.) (14 times)
- Speaking at domestic and international conferences (75 times)
- Dissemination within the Ministry of Foreign Affairs (to improve science and technology literacy)

Networking with domestic and international science and technology professionals (Science for Diplomacy)

- Launch and participation in FMSTAN (Foreign Ministries Science and Technology Advice Network)
- Exchange of views with dignitaries from various countries (including ministerial-level dignitaries)
- Promotion of cooperation with domestic science and technology related parties (Ministries, Agencies, and Corporations)

In promoting the above, Mr. Kishi visited 23 countries, exchanged

views with science and technology executives and researchers more than 100 times in Japan and abroad, and spoke at related symposiums and international conferences more than 70 times. In addition, at the 2nd Symposium on Science and Technology Diplomacy organized in December 2019, he proposed the following as the direction of future science and technology diplomacy.

- (1) In the midst of the advancing digital revolution, while science and technology diplomacy based on scientific evidence data has been promoted in the past, the promotion of science and technology diplomacy with data science at its core, including rule formation, should be strengthened more than ever in the future. Science and technology diplomacy must contribute to the realization of Society 5.0 and the achievement of the SDGs, avoiding a negative social direction due to data monopolies and digital despotism.
- (2) However, it is also necessary to distinguish between “science diplomacy” and “technology diplomacy,” bearing in mind the rise of emerging technologies in security (global security) and the existence of countries and regions that do not necessarily share the same ethical values in science. It is also necessary to distinguish between “science diplomacy” and “technology diplomacy”.
- (3) Science and technology capability is the greatest tool for Japan’s science and technology diplomacy, and the research system should be strengthened by considering the “Japanese model” rather than simply imitating Western systems. Similarly, in promoting science and technology diplomacy, a medium- to long-term vision should be strategically formulated based on the “Japanese model,” and this vision should be widely disseminated both domestically and internationally.
- (4) In order to put the above into practice, it is essential that appropriate budgetary measures be taken. In addition, it is necessary to strengthen cooperation with various sectors, such as related ministries, research institutes, universities, and industries, and to develop human resources. Furthermore, cooperation within the Ministry of Foreign Affairs should be strengthened with related departments and diplomatic missions abroad, and consideration should be given to increasing the number of science and technology attachés in diplomatic missions abroad.

2.2 United States

In 2000, then U.S. Secretary of State Madeleine Albright pioneered the establishment of a Science and Technology Advisor position to the Secretary of State and a supporting Office of the Science and Technology Adviser. This was in response to a 1999 report by the National Research Council (NRC) that recommended “the creation of a senior advisor to the Secretary of State to provide direct advice on matters related to science, technology, and health care,” and the establishment of an Office of the Science and Technology Adviser to the Secretary of State to provide direct advice to the Secretary of State. The report also responded to the enactment of federal legislation requiring the Secretary of State to have a Science and Technology Advisor who advises the Secretary of State directly. Former Science and Technology Advisors to the Secretary of State are shown in Figure 4.

Figure 4 Science and Technology Advisor to successive Secretaries of State

Term (years)	Name	Special field	Main previous jobs
2000 – 2003	Norman Neureiter	Organic chemistry	Scientific Attaché in Warsaw and Bonn Vice President, Texas Instruments. Assistant to the Nixon Administration’s Science Advisor for International Affairs.
2003 – 2007	George Atkinson	Photochemistry	Science Fellow, American Physical Society Professor, University of Arizona
2007 – 2010	Nina Fedoroff	Molecular biology	Professor, Pennsylvania State University
2011 – 2014	William Colglazier	Theoretical physics	Chief Operating Officer of the National Research Council and the National Academy of Sciences. Assistant to the Congressman by the National Association for the Advancement of Science (AAAS) Congressional Science Fellowship, Professor at the University of Tennessee, and Editor-in-Chief of Science & Diplomacy magazine
2015 – 2017	Vaughan Turekian	Environmental science atmospheric chemistry	Chief International Officer, AAAS Director, AAAS Center for Science and Technology Diplomacy Special Assistant to the Under Secretary for International Affairs, Department of State
2019 – 2020	Mung Chiang	Electrical engineering mathematics	Professor of Engineering, Princeton University Dean of Engineering, Purdue University
2023 – 2025	Patricia Gruber	Ocean physics	Director of Research, Office of Naval Research (ONR)

* Mr. Gruber resigned in January 2025, and as of February 2025, State Department employee Allison Schwier is acting Science and Technology Advisor to the Secretary of State. State Department staff (Frances Colón, Matt Chessen, and Allison Schwier, respectively) served as Acting Science and Technology Advisor to the Secretary of State in 2014-2015, 2017-2019, and 2021-2023.

Prepared by CRDS based on various sources

Colglazier, the fourth Science and Technology Advisor to the Secretary of State, described the role of the Secretary of State’s Office of Science and Technology Advisor (STAS), saying, “It assists the State Department, but it is not an operating bureau. Its goal is to strengthen the science and technology knowledge base within the department, anticipate science and technology issues that may affect foreign policy, advocate science-based decisions in all countries, and support global science and technology engagement that serves U.S. interests.”¹⁸ Patricia Gruber was appointed Science Advisor to the Secretary of State in September 2023, but resigned in January 2025, and as of February 2025, Allison Schwier, a State Department employee, is the Acting Advisor. Immediately after her appointment, in November 2023, Ms. Gruber became Chair of the Foreign Ministries Science and Technology Advice Network (FMSTAN) (described in detail in Chapter 2, 3.1), where she worked to strengthen the network of science advisors in each country.

The U.S. State Department’s major initiatives include the U.S. Science Envoy Program and the Embassy Science Fellows Program. The former, launched in 2010, appoints six to seven eminent scientists as Science Envoys to promote international cooperation, expand human resource networks, and solve common problems by holding conferences on topics related to foreign policy and science and technology innovation, and by visiting other countries. Approximately 30 people were appointed by 2022, and seven new appointments were made in 2023¹⁹. The latter program sends an average of 50 scientists to U.S. diplomatic missions abroad each year. Since its inception in 2001, approximately 550 fellows have participated in this program, contributing to diplomatic activities in the field of science and technology²⁰. In January 2023, the Office of the Special Envoy was also established to formulate and coordinate foreign policy on important emerging technologies such as artificial intelligence (AI), quantum information science, and biotechnology.

In addition, the American Association for the Advancement of Science (AAAS) has a significant presence in U.S. science and technology diplomacy. The AAAS established the Center for Science Diplomacy in 2008, and in 2010, in collaboration with The Royal Society, advocated the three elements of current science and technology diplomacy. In addition, the AAAS publishes the journal *Science Diplomacy*²¹, provides training courses for developing countries in collaboration with the World Academy of Sciences (TWAS), is a leading advocate of the UN STI for SDGs Forum, and is involved in the World Science Forum (WSF). Most recently, together with The Royal Society, it published a new report, “Science diplomacy in an era of disruption” in February 2025 (see Chapter 1, 1.2). The AAAS also takes advantage of its non-governmental

18 Science, “Science and Diplomacy,” <https://www.science.org/doi/full/10.1126/science.1220355>
19 DoS, “Announcement of New Cohort of U.S. Science Envoys,” <https://www.state.gov/announcement-of-new-cohort-of-u-s-science-envoys/>
20 DoS, “Embassy Science Fellows Program,” <https://www.state.gov/programs-office-of-science-and-technology-cooperation/embassy-science-fellowsprogram/>
21 AAAS, “Science & Diplomacy,” <https://www.sciencediplomacy.org/>

status to promote scientific collaboration with Cuba, the former Soviet Union²², North Korea²³, China²⁴, and other countries (AAAS activities are detailed in Chapter 3, 3.3.).

2.3 United Kingdom

In 2009, the United Kingdom appointed theoretical chemist David Clary, a professor at Oxford University, as the first Principal Scientific Advisor to the Foreign and Commonwealth Office.²⁵ He was succeeded by materials physicist Robin Grimes, Professor of Materials Physics at Imperial College, from 2013-2018²⁶. The two actively interacted with and contributed to various diplomacy efforts, including Japan, at the dawn of science and technology diplomacy. As of February 2025, Charlotte Watts, Professor in Social and Mathematical Epidemiology at the London School of Hygiene and Tropical Medicine, is the UK Foreign Office's Chief Science Advisor. In addition to the Chief Science Advisor, the UK has established science advisors in the Ministry of Foreign Affairs and other ministries, working closely with them. Although science and technology diplomacy is led by the Principal Advisor to the Ministry of Foreign Affairs, Angela McLean, Chief Scientific Adviser to the Government, who took office in April 2023, is also engaged in promoting international collaboration and building international relations.

In its "Integrated Review of Security, Defence, Development, and Foreign Policy (2021)"²⁷, the UK government states that it will position science as the foundation of the UK's ability to address societal challenges and boost prosperity, and maintain the UK's position at the cutting edge of science and technology, and that collective action with allies and partners and co-creation will be crucial over the next decade. In November 2022, the UK Parliamentary Office of Science and Technology updated its 2018 policy paper on science and technology diplomacy²⁸, stating that addressing technological change and new geopolitical tensions also requires innovative diplomatic approaches, and that, for example, some academia are seeking to work with partners in authoritarian regimes with clear they are calling for the definition of rules and instruments, stating.

- 22 AAAS, "Event | U.S.-Soviet Scientific Cooperation: A Look at the History and Lessons for Science Engagement for the Future," <https://www.aaas.org/events/event-us-soviet-scientific-cooperation-look-history-and-lessons-scienceengagement-future>
- 23 AAAS, "Event | A Peace of Science: Diplomacy with North Korea," <https://www.aaas.org/events/event-peace-science-diplomacy-north-korea-0>
- 24 AAAS, "Connecting Scientific Communities: AAAS Travels to China," <https://www.aaas.org/news/connecting-scientific-communities-aaas-travels-china>
- 25 The Royal Society, "David Clary," <https://royalsociety.org/people/david-clary-11232/>
- 26 GOV.UK, "New Chief Scientific Adviser for the FCO – Prof Robin Grimes appointed," <https://www.gov.uk/government/news/new-chief-scientific-adviser-for-the-fco-prof-robin-grimes-appointed>
- 27 GOV.UK, "Global Britain in a Competitive Age: the Integrated Review of Security, Defence, Development and Foreign Policy," <https://www.gov.uk/government/publications/global-britain-in-a-competitive-age-the-integrated-review-ofsecurity-defence-development-and-foreign-policy>
- 28 UK Parliament, "Science Diplomacy," <https://post.parliament.uk/science-diplomacy/>

2.4 EU

In the EU, the term “science diplomacy” first appeared in a 2012 report from the European Commission to European governments entitled “Strengthening and focusing EU international cooperation in research and innovation: a strategic approach”. The report states that science diplomacy “will revitalize international cooperation in research and innovation as a soft power instrument and as a mechanism to improve relations with key countries and regions”. This concept was further developed in the 2016 report “Open Innovation, Open Science, Open to the World: a European Vision”, which calls for “international research leading to common standards, exchange and mobility of science, sharing of resources and facilities, scientific advice to diplomats and diplomatic scientists, and innovation cooperation to support good governance and policymaking and to help build mutual relationships”. In 2017, the European Commission concluded through its study “Tools for EU Science Diplomacy” that science diplomacy efforts in Europe remain largely uncoordinated, and that diplomatic and security policies, and an appropriate EU Science Diplomacy Strategy and related action plan were needed to support it. In response, the European Commission implemented three research projects under the Horizon 2020 program.

- Using Science for/in Diplomacy for Addressing Global Challenges (S4D4C) (2018-2021)
- Inventing a Shared Science Diplomacy for Europe (InsSciDE) (2018-2021)
- European Leadership in Cultural, Science and Innovation Diplomacy (EL-CSID) (2016-2019)

All of this has led to a much better understanding of European science diplomacy, the consideration of options, the development of training materials and statements such as the Madrid Declaration on Science Diplomacy, and the creation of The European Union Science Diplomacy Alliance²⁹. Science and technology diplomacy efforts are also underway at the regional and local level, as evidenced by the creation of the SciTech DiploHub by the city of Barcelona.

In May 2021, the European Commission presented a new strategy for international cooperation in science and technology, the “Global Approach to Research and Innovation”, to jointly tackle global challenges in areas such as climate change, digital transition, and health. This reaffirmed the EU’s commitment to cooperation based on common values. The strategy calls for reciprocity and a level playing field as a basis for building trust with international partners and stresses “the need to strengthen the emphasis on science and technology in the EU’s foreign and security policy in the context of ‘science diplomacy’”. The Council then asked the European Commission and the European External Action Service “to prepare and submit to the Council a European Science Diplomacy Agenda; to consider designating a science contact point to ensure that EU delegations have adequate science competence; and to promote cooperation with third country science advisors in member states”. In response, the European

²⁹ EU Science Diplomacy Alliance, “The European Union Science Diplomacy Alliance,” <https://www.science-diplomacy.eu/>

Commission established a Working Group on Science Diplomacy in November 2023 with the aim of developing potential elements of a future European science diplomacy framework, and is currently studying the formation of a “European Science Diplomacy Framework (tentative name)”. As part of these studies, Spain, which holds the Presidency of the Council of Europe, and the European Commission jointly held the 1st European Science Diplomacy Conference in Madrid in December 2023, which provided one direction for future international discussions on science diplomacy, emphasizing hard power as well as soft power as the objective of science diplomacy³⁰. In February 2025, an expert report entitled “A European Framework for Science Diplomacy” was published. It also stated that the 2nd European Conference on Science Diplomacy will be held in 2025.

Column 3

First European Conference on Science Diplomacy (December 2023 in Madrid) Outcome Document

** Summary of documents published on the website by CRDS*

Given the rapidly evolving geopolitical and technological landscape, Europe needs to act strategically and develop an ambitious European framework for science diplomacy.

- This framework should build on the impressive legacy of European science diplomacy and the number of existing activities and initiatives at the EU level and in Member States.
- The framework should draw on the expertise and skills of the rapidly growing community of scholars and practitioners of European science diplomacy.
- The framework should identify synergies and maximize the EU’s added value from joint action through appropriate linkages between research and foreign policy instruments and the creation of space for exchange between the scientific and diplomatic communities.
- The framework should serve to define the principles that should form the basis for European science diplomacy action, based on the EU’s global approach to research and innovation.
- This framework should rethink European science diplomacy in a fragmented and multipolar world and reflect the use of science

30 EU, “European Science Diplomacy Conference,” <https://eu-science-diplomacy.service-facility.eu/>

diplomacy as soft and hard power, based on the concept of being as open as possible but closed as necessary.

- The framework needs to reflect how European diplomacy can be more strategic, effective, and resilient by leveraging scientific evidence and foresight, including enabling diplomacy to address the opportunities and risks of disruptive technologies.
- The framework should promote the development of science-based solutions to address the growing pressures by public and private actors on the global commons, including the Arctic, oceans, and low orbit, and support rules-based multilateralism.
- The framework must leverage the impact of research and innovation to address systemic global challenges such as climate change, biodiversity loss, and environmental pollution to achieve the Sustainable Development Goals and thereby create better lives for all.
- The framework should strengthen the role of science diplomacy in the diplomatic missions of the EU and its Member States, including through science advisors, and promote the EU's global science diplomacy activities, for example by engaging with diaspora scientists and graduates of migration programs.
- This framework would build on the successful example of the Partnership for Research and Innovation in the Mediterranean Area (PRIMA) to promote relations with the European neighborhood as well as with strategic partners in other world regions, such as Latin America and the Caribbean. It will also promote the development of new partnerships in the Mediterranean region, such as the European Union Africa, and the Indo-Pacific.
- The framework should serve to promote capacity building in European science diplomacy through the training of scientists, diplomats, and their counterparts, and through advancing the frontiers of science diplomacy through interdisciplinary research, including in the social sciences and humanities.

2.5 Global South

2.5.1 General remarks

There is no clear definition or list of nations comprising the “Global South”, but the term is generally used to refer to developing countries (due to the fact that most developing countries are located in the Southern Hemisphere), and is used in this paper to refer to nations such as

those found in the G77, a UN negotiating group³¹.

The G7 and other Western industrialized countries have begun to take a more concrete stance toward countries in the Global South, whose position in the international community and economic power are increasing, to strengthen cooperation. The OECD held a Ministerial Council meeting in June 2023 (chaired by the UK) under the theme “Securing a Resilient Future: Common Values and Global Partnerships” and adopted the “OECD Indo-Pacific Strategic Framework”. The Framework positions the Indo-Pacific region as important in terms of the supply chain for semiconductors and clean energy technologies, and emphasizes the strengthening of cooperation between the Indo-Pacific countries and ASEAN and the OECD. In addition, China, the U.S., the U.K., Germany, and other countries where competition for scientific and technological human resources is intensifying internationally are taking measures such as easing visa and nationality requirements to attract excellent research personnel from the Global South. At the same time, many of the countries positioned in the Global South are promoting their own foreign policies to solve their own economic and social problems, and there are also moves among the Global South to strengthen bilateral or multi-country partnerships. For example, in January 2023, India held a summit and ministerial-level meeting titled “Voice of Global South Summit”³² to bring together the countries of the Global South, and said that the views expressed at the meeting would be reflected in the G20 discussions. In 2023, India held the G20 Presidency, and the African Union (AU) was invited to become a permanent member³³. The “G20 New Delhi Summit Declaration” expressed support for stronger relations between the G20 and the AU, support for African industrialization, and support for the AU’s development goal “Agenda 2063”³⁴. Also, at the G7 Puglia Summit in June 2024, representatives of Canada, the Republic of Congo, Cote d’Ivoire, Ethiopia, France, Germany, Italy, Japan, Kenya, Mozambique, Nigeria, South Africa, the United Kingdom, the United States, and the European Union adopted a joint statement on the initiative “Energy for Growth in Africa”³⁵.

Against this backdrop, Japan has been taking concrete steps to strengthen its partnership with the Global South³⁶. Japan has provided a high level of support to the Global South through ODA and contributions to international development finance institutions (Japan’s Official Development Assistance (ODA) in 2021 was the third largest among developed countries,

31 United Nations, “The Group of 77 at Fifty,” <https://www.un.org/en/chronicle/article/group-77-fifty>

32 Ministry of External Affairs, Government of India “1st Voice of Global South Summit 2023,” <https://www.mea.gov.in/voice-of-global-summit.htm>

33 “African Union appeals to G20 members for Modi’s support; may it come true next year?” <https://jp.reuters.com/world/us/RWATIULBG5IUHCH4SIKWWHFXAA-2023-09-07/>

34 “G20 summit agrees to permanent membership of African Union (AU)” <https://www.jetro.go.jp/biznews/2023/09/8e0f4f9044916a5a.html>

35 Ministry of Foreign Affairs, “G7 Puglia Summit” https://www.mofa.go.jp/mofaj/ecm/ec/pageit_000001_00005.html

36 Prime Minister’s Office, “Conference on Promoting Strengthening Cooperation with Global South Countries” https://www.kantei.go.jp/jp/101_kishida/actions/202310/17globalsouth.html

following the United States and Germany³⁷). The term “Global South” was included for the first time in the 2022 edition of the White Paper on Development Cooperation, which expressed the intention to strengthen emergency assistance, including food aid, to Global South countries in Asia, Africa, and the Middle East that are facing food insecurity and energy crises in the wake of the situation in Ukraine. The white Paper also stressed the need for Japan to support the Global South as a measure to maintain and strengthen a peaceful and stable international order³⁸. In August 2025, Japan will host The Ninth Tokyo International Conference on African Development (TICAD 9)³⁹ in Yokohama.

This section discusses the partnerships with ASEAN, which has been increasing its presence in the Global South in recent years. Since 2021, the U.S., EU, and China have been strengthening their government-level partnerships with ASEAN and their financial support. These partnerships include enhanced collaboration in the areas of science and technology, and a series of plans have been announced to provide assistance for environmental issues, supply chain, defense, and development.

In addition, the United Kingdom has maintained a government-level partnership with ASEAN since leaving the EU, and in January 2024, announced the establishment of the International Science Partnerships Fund (ISPF), which will be funded by the ODA, and will support collaboration with global researchers and innovators on global environment, health, technology, and human resources for Southeast Asian countries. The Fund is funded by a consortium of leading research and innovation organizations in the UK, including UK Research and Innovation (UKRI) and UK Academies, and is intended to support collaboration with researchers and others in Southeast Asian countries from around the world. It has been announced that Malaysia will be the first recipient of this fund⁴⁰.

The year 2023 marks the 50th anniversary of friendly and cooperative relations between Japan and ASEAN. Japan and ASEAN have built up their relationship over a long period of time and have made steady progress in specific cooperation projects in a wide range of areas such as the economy, security, researcher exchange, etc. In the STI field they have conducted joint research to solve social issues, human resource development utilizing Japanese technology and knowledge, as well as exchanges with Japanese youth in the field of science and technology⁴¹.

37 Ministry of Foreign Affairs, “Japan’s Development Cooperation White Paper 2022 Introduction” https://www.mofa.go.jp/mofaj/gaiko/oda/shiryo/hakusyo/22_hakusho/honbun/b0/hajimeni.html

38 Ministry of Foreign Affairs, “Japan’s Development Cooperation White Paper 2022 Assistance to the Global South, which will be greatly affected” https://www.mofa.go.jp/mofaj/gaiko/oda/shiryo/hakusyo/22_hakusho/honbun/b1/s1_3.html

39 Ministry of Foreign Affairs, “The 9th Tokyo International Conference on African Development (TICAD 9)” https://www.mofa.go.jp/mofaj/af/af1/pagew_000001_00002.html

40 GOV.UK, “UK launches International Science Partnership Fund worth £218 million,” <https://www.gov.uk/government/news/uk-launches-international-science-partnership-fund-worth-218-million>

41 Ministry of Foreign Affairs, “The 50th Anniversary of ASEAN-Japan Friendship and Cooperation ~ People-to-People Interaction ~” <https://www.mofa.go.jp/mofaj/files/100596241.pdf>

Against this backdrop, the “Special Summit Meeting to Commemorate the 50th Anniversary of ASEAN-Japan Friendship and Cooperation” was held in Tokyo in December 2023. At the Summit, a joint vision statement, the “Actions for ‘Peace and Prosperity’ through ‘Co-Creation’ Based on Trust⁴²”, was adopted to strengthen the mutually beneficial comprehensive strategic partnership between Japan and ASEAN. The Actions include a total of 130 specific cooperation items, including plans to further promote science and technology cooperation by strengthening cooperation in research and development and innovation, and to foster the exchange of human resources between ASEAN and Japan. In addition, the FY2023 supplementary budget provided for “ASEAN-Japan Cooperative Research Collaboration in Science, Technology and Innovation (approximately 14.6 billion yen)” to strengthen sustainable research collaboration with ASEAN countries through a wide range of initiatives building on the longstanding ASEAN-Japan international joint research and research human resource exchange.

Furthermore, in April 2024, MEXT, the ASEAN Secretariat, and JST organized the “Japan ASEAN High-level Roundtable on Science, Technology and Innovation⁴³” as an official side event of the ministerial-level meeting of the Committee on Science and Technology Policy (CSTP) of the OECD, which was attended by science, technology, and innovation (STI) policy officials from ASEAN member countries. STI policy leaders from Japan and ASEAN member countries, including science and technology ministers, science and technology advisors, and government officials, gathered to discuss ASEAN-Japan STI cooperation. The participants confirmed that Japan, ASEAN, and ASEAN member countries have been building important partnerships. They also welcomed the opportunity provided by the 50th anniversary of friendship and cooperation in 2023 to further deepen STI cooperation in the future.

2.5.2 Science and Technology Diplomacy Initiatives

Since the concept of science and technology diplomacy was first proposed by the United States and the United Kingdom in 2010, science and technology diplomacy has been promoted mainly by developed countries. In recent years, science and technology diplomacy efforts by the Global South have become more apparent, as evidenced by the G20 Science Advisory Roundtable held in 2023 under the leadership of India, which called for the “establishment of a global science advisory mechanism using science diplomacy as a tool” (See Chapter 3, 3.1). The International Network for Governmental Science Advice (INGSA) also held an international conference in Rwanda in May 2024, the first time an INGSA international conference has been held in the Global South, that was also attended by participants from Africa, and at which science and technology diplomacy was discussed (see Chapter 3, 3.3.1 for more details). In addition, the 11th World Science Forum, held in November 2024, was relatively well attended by participants from Europe and Africa and also addressed science and technology diplomacy (detailed in

⁴² Ministry of Foreign Affairs, “The 50th Anniversary of ASEAN-Japan Friendship and Cooperation/ Outcome of the Special Summit” <https://www.mofa.go.jp/mofaj/files/100597146.pdf>

⁴³ JST, “ASEAN-Japan High-Level Roundtable on Science, Technology and Innovation Held” <https://www.jst.go.jp/report/2024/240621.html>

Chapter 3, 3.3.5).

Asian trends include the efforts of the Asia-Europe Foundation (ASEF) and the Thailand-led ASEAN+3 (Japan, China, and Korea) Pilot Science Diplomacy Workshop for Young Mid-career Researchers.

Initiatives of the Asia-Europe Foundation (ASEF)

The Asia-Europe Foundation (ASEF) was established at the 1st Asia-Europe Meeting (ASEM) to promote exchanges between the two regions, and was launched on the occasion of the First ASEM Foreign Ministers’ Meeting in February 1997. With its secretariat in Singapore, it is operated by 21 Asian countries and one organization, and 30 European countries and one organization, all of which are ASEM member countries. Japan is also member. ASEF conducts a variety of activities, including personal, intellectual, and cultural exchanges, to promote mutual understanding between Asia and Europe. Beginning in 2024, the Asia-Europe Science and Technology Diplomacy Report was launched with the aim of identifying differences in national strategies and policy measures related to science and technology diplomacy among member countries and regions, and providing insights and recommendations to relevant organizations on areas for strengthening intra-regional and inter-regional cooperation. Country informants (country researchers) were publicly invited from about 40 countries and regions to participate in the preparation of the report, which was scheduled to be released in March 2025 or later. The JST Center for Research and Development Strategy (CRDS) Science and Technology Diplomacy Group participated from Japan. In November 2024, as a side event of the 11th World Science Forum, an interim report meeting was held to provide an overview from France, Belgium, Hungary, Vietnam, Thailand, and Japan. Although none of the countries that presented a topic this time has a national strategy dedicated to “science and technology diplomacy”, they did tend to emphasize science and technology diplomacy in their STI policies. France, Belgium, and Hungary emphasized that they will promote their own efforts in accordance with the “European Science Diplomacy Framework (tentative name)” currently being formulated by the EU Directorate-General for Research. Vietnam and Thailand have included science and technology diplomacy in their policy documents, indicating a growing momentum to promote specific initiatives in the future. As for Japan, SATRESP, ASPIRE, and NEXUS were introduced as examples of initiatives utilizing the Science and Technology Advisory System and in line with the Science and Technology Innovation Basic Plan. Discussions at the 3rd Symposium on Science and Technology Diplomacy were also introduced, and recommendations were made to move from policy to practice (Actionable Science and Technology Diplomacy).

The Pilot Science Diplomacy Workshop for Early-to-Mid Career Researchers in ASEAN Plus Three (Japan, China, and Korea)

This workshop was held in August 2024 in Bangkok, Thailand, hosted by the Thai Science Research and Innovation Organization (TSRI) and co-sponsored by the Third World Academy of Sciences (TWAS), the American Association for the Advancement of Science (AAAS), and the National Science Museum (NSM) of Thailand. Modeled on the AAAS-TWAS Summer Course on Science Diplomacy, which has been held annually by TWAS and AAAS in Trieste, Italy since

2014, the workshop aimed to foster the concept of science and technology diplomacy in the ASEAN region and to build a network of young and mid-career researchers and government officials. Pairs of researchers and administrative officials from nine ASEAN countries (excluding Singapore) and Japan participated. The keynote session featured a speech by the Japanese Ambassador to Thailand, as well as online participation by the Science and Technology Advisor to the U.S. Department of State and the International Director of AAAS. In addition, the TWAS Science Diplomacy Program Coordinator spoke on the basics of science diplomacy, and experts such as Zakri Abdul Hamid, former Science Advisor to the Government of Malaysia, presented on related topics. From Japan, the CRDS Science and Technology Diplomacy Group introduced the trends and directions of science diplomacy discussions in the world, including the 3rd Science and Technology Diplomacy Symposium, and noted the growing importance of research security. Group work and presentations were then conducted by the participants. It appears that consideration is being given to holding the next meeting, and it is hoped that the event will be held on an ongoing basis.

3 | STI Issues Surrounding Science and Technology Diplomacy

3.1 Science and Technology Diplomacy and the Scientific Advisory System

The importance of scientific and technical advisors and scientific advice⁴⁴ has received renewed attention. COVID-19 was truly a global pandemic in human history, but one of the few positive aspects was that in overcoming it, much of the public came to understand the importance of science, technology, and scientific advice. Furthermore, the rapid development of emerging technologies such as AI, quantum, and biotechnology, combined with changes in world affairs due to geopolitical tensions, are rapidly increasing the demand for scientific knowledge in important policy decisions.

Discussions on Japan's science and technology advisory system began in light of how the science and technology advisory systems in the United Kingdom and the United States provided scientific advice to the Japanese government in the aftermath of the Great East Japan Earthquake and Fukushima Daiichi Nuclear Power Plant accident in March 2011. At the time of the Great East Japan Earthquake, Sir John Beddington, then Chief Scientific Advisor to the British government, immediately convened an emergency group called SAGE (Scientific Advisory Group for Emergencies), which issued guidelines based on scientific evidence to embassy staff in Japan and the British public that concluded "there is no need to evacuate Tokyo". A few days after the earthquake, when the U.S. Navy was advocating the evacuation of U.S. citizens from the Tokyo area and the State Department was opposing the evacuation from a diplomatic standpoint, John Holdren, then Assistant to the President for Science and Technology, advised the President not to evacuate based on simulations supported by the scientific community in the United States. This decision became the final decision of the U.S. government⁴⁵. These actions led to a discussion on whether Japan should also have its own science and technology advisor to provide scientific advice to the government.

In September 2015, the Ministry of Foreign Affairs appointed Teruo Kishi (Professor Emeritus, University of Tokyo) as the first Science and Technology Advisor to the Minister of Foreign Affairs, based on the recommendations of the Advisory Panel on Science and Technology Diplomacy (chaired by Takashi Shiraishi, then President of National Graduate Institute for Policy Studies). In April 2020, Yoichiro Matsumoto (Professor Emeritus, University of Tokyo) became

⁴⁴ For more information on scientific advice, see "Takeo Arimoto, Yasushi Sato, and Keiko Matsuo 2016, Scientific Advice: Science, Technology, and Policy Formation in the 21st Century," University of Tokyo Press, and "Toward the Establishment of an Ecosystem of Scientific Advice in the Post-Pandemic Era: Challenges and Future Directions in Response to New Coronavirus Infections. Challenges and Future Directions in Responding to Viral Infections: <https://www.jst.go.jp/crds/pdf/2022/RR/CRDS-FY2021-RR-10.pdf>" etc.

⁴⁵ Nature.com, "Five years after Fukushima: scientific advice in Japan," <https://www.nature.com/articles/palcomms201625>

the second Science and Technology Advisor to the Minister of Foreign Affairs. In the meantime, Mitsunobu Kano (Professor, Okayama University) served as Deputy Science and Technology Advisor to the Minister of Foreign Affairs from 2019 to 2022, and was succeeded by Motoko Kotani (Professor, Tohoku University) in 2022. (The activities of the Science and Technology Advisor to the Minister of Foreign Affairs are detailed in Chapter 2, 2.1.)

In September 2022, the Japanese government appointed Kazuhito Hashimoto (Professor Emeritus of the University of Tokyo and President of JST) to the position of Science and Technology Advisor to the Cabinet Secretariat, a position equivalent to that of the Presidential Science and Technology Advisor in the United States or the Chief Science Advisor to the Government in the UK. This appointment was made in response to the “Grand Design for New Capitalism and Action Plan (Cabinet Decision: June 7, 2022)⁴⁶”, which included “the establishment of a Science and Technology Advisor at the Prime Minister’s Office to provide information and advice to the Prime Minister.” In May 2023, Mr. Shigeo Koyasu (President, Research Organization of Quantum Science and Technology) was appointed Science and Technology Advisor to the Minister of Education, Culture, Sports, Science and Technology, and in March 2024, Mr. Hideo Ohno (then President, Tohoku University) was appointed as Special Advisor (Science and Technology) to METI. As of February 2025, four science and technology advisors to the government are active and collaborating with each other by holding liaison meetings, etc.

Cabinet Secretariat Science and Technology Advisor Hashimoto is engaged in a variety of activities aimed at gathering information and building a network that will contribute scientific advice to the Prime Minister. In March 2023, prior to the G7 Hiroshima Summit, the G7 Science and Technology Advisors Roundtable⁴⁷ was held in Washington, D.C., for a frank discussion on the promotion of international brain circulation and the associated need for research integrity and security. In addition, the Scientific Advisory Roundtable 2023⁴⁸ and the Scientific Advisory Roundtable 2024⁴⁹ were held in Kyoto in September 2022 and October 2023, respectively, to exchange views on the expected roles and initiatives for scientific and technical advisors and advice. The Scientific Advisory Roundtable 2024 took advantage of the occasion of the 21st Annual Meeting of the International Forum on Science, Technology and the Future of Humanity (STS Forum) to hold a frank discussion among domestic and international science and technology advisors and equivalent experts attending the meeting on the role of scientific and technical advisors and scientific advice for ensuring research security. Furthermore, Advisor Hashimoto, together with Takahiro Ueyama, Full-time Member of the Council for Science, Technology and Innovation, and Shigeo Koyasu, Science and Technology Advisor to the Minister

⁴⁶ Cabinet Office, “Grand design and implementation plan for the new capitalism” https://www.cas.go.jp/jp/seisaku/atarashii_sihonsyugi/pdf/ap2022.pdf

⁴⁷ Cabinet Office, “G7 Science and Technology Advisors Roundtable” https://www.cas.go.jp/jp/siryu/pdf/20230324_kanrisyoku.pdf

⁴⁸ JST, “Science Advisor Roundtable 2023 to be held” <https://www.jst.go.jp/report/2023/231121.html>

⁴⁹ JST, “Scientific Advice Roundtable 2024 to be held” <https://www.jst.go.jp/report/2024/241108.html>

of Education, Culture, Sports, Science and Technology, jointly with the ASEAN Secretariat, organized the “ASEAN-Japan Science, Technology and Innovation High Level Roundtable⁵⁰” as an official side event at the OECD Headquarters (Paris, France) in April 2024, taking advantage of the opportunity provided by the ministerial-level meeting of the Committee on Science and Technology Policy (CSTP) of the OECD to bring together science, technology, and innovation (STI) policy makers from ASEAN member countries.

Science and technology advisors are also becoming more active in the international community. In August 2024, the Chief Science Advisors of Australia, Canada, New Zealand, the United Kingdom, and the United States issued a joint statement on their shared commitment to using science and technology to address today’s challenges and improve collaboration across the scientific research systems within and between their countries⁵¹. One sign of the growing interest in scientific advice in emerging economies are the G20 activities. The G20 presidency in 2023 was India. At the initiative of Ajay Kumar Sood, Chief Science Advisor to the Government of India, the G20 Chief Science Advisers Roundtable was held twice (March and August) and an outcome document⁵² was compiled that called for “the establishment of a global science advisory mechanism using science diplomacy as a tool”. This document was adopted as an annex to the G20 Summit in September 2023. In September 2024, UNESCO and the Government of India co-hosted the Chief Science Advisers Roundtable 2024 at UNESCO Paris Headquarters with the participation of national chief science advisers or their equivalent and international organizations. The discussion focused on “Promoting Open Science, Eliminating Knowledge Asymmetries, and Building Scientific Advisory Capacity Globally”⁵³.

In the second Trump administration in the United States, President Trump nominated Michael Kratsios, an executive at AI startup Scale AI, as Director of the White House Office of Science and Technology Policy (OSTP) and Assistant to the President for Science and Technology (APST), who will also serve as science and technology advisor to the President. On January 23, President Trump also announced the establishment of the President’s Council of Advisors on Science and Technology (PCAST) under his administration, and nominated David Sacks, a former executive at online payments giant PayPal Inc., as its head. President Trump also created two new posts of senior policy advisor and counsel for AI at OSTP and appointed Sriram Krishnan, an Indian-origin investor, and Lynne Parker, an AI researcher, respectively, for each post. Parker will also

50 JST, “ASEAN-Japan High-Level Roundtable on Science, Technology and Innovation Held”
<https://www.jst.go.jp/report/2024/240621.html>

51 British Government, “Joint Statement by the United Kingdom, Australia, Canada, New Zealand, and the United States: August 2024”
<https://www.gov.uk/government/publications/joint-statement-between-the-uk-australia-canada-new-zealandand-the-united-states-august-2024>

52 G20 2023 INDIA, “G20- Chief Science Advisers Roundtable (G20-CSAR) Outcome Document & Chair’s Summary,”
https://www.g20.in/content/dam/gtwenty/gtwenty_new/document/G20_CSAR_Outcome_Document_and_Chair_Summary_28Aug.pdf

53 UNESCO Digital Library “Chief Science Advisers Roundtable 2024 (CSAR-2024)”
<https://unesdoc.unesco.org/ark:/48223/pf0000391404>

serve as Executive Director of PCAST. Considering that in the first Trump administration, the OSTP Director was appointed approximately two years after the start of the administration, and PCAST was launched two years and nine months later, the increased interest in science and technology in the second administration is evident.

3.2 Science and Technology Diplomacy and Research Security

3.2.1 The Changing Positions of Science and Technology and STI Policies in the Midst of the Changing Global Situation

In 1999, the Budapest Declaration⁵⁴ was adopted at the World Science Forum (WSF) (see Chapter 3, 3.3.5), and “science for society” became a common understanding. In the 2000s, expectations for economic development through science, technology, and innovation (STI) increased, while the Sustainable Development Goals (SDGs) were adopted at the UN Summit in 2015, and momentum for solving global-scale issues through STI also accelerated. However, the role of STI is changing even further. In recent years, economic security challenges have emerged in the wake of COVID 19, which developed into an unprecedented pandemic, and growing supply chain risks due to heightened geopolitical tensions. In response, the United States, the United Kingdom, Germany, and the European Union are actively participating in a framework of like-minded countries and are revising their national security strategies to emphasize science and technology. Recent national and regional national security strategies and economic security policies (Figure 6) show the growing involvement of science and technology (especially emerging and critical technologies), as well as advocating for the “formation of an international order” that is free and open with a “multinational approach”. In other words, in order to develop and socially implement emerging technologies (AI, semiconductors, quantum, advanced telecommunications, biotechnology, clean energy, etc.) and secure supply chains, Japan will need to focus on establishing its technological and industrial superiority and working with allies that share its values, while also attempting to solve global-scale issues such as climate change. The strategy of simultaneously implementing these conflicting approaches can be described as a “dual-track approach”.

Figure 6⁵⁵ National security strategies and economic security policies of major countries and regions

Date of Establishment	Document Name	Decision-making body
October 2022	US National Security Strategy	Whitehouse

⁵⁴ World Science Forum (1999). “DECLARATION ON SCIENCE AND THE USE OF SCIENTIFIC KNOWLEDGE,” n.d. <https://2019.worldscienceforum.org/contents/declaration-of-the-1999-unesco-world-conference-on-science-110056>.

⁵⁵ Figure 5 is included only in the Japanese version.

Published 2021, revised March 2011	UK Integration Review	British Cabinet Office
June 2023	German National Security Strategy	German Federal Government
June 2023	European Economic Security Strategy	European Commission (executive of the EU)

Prepared by CRDS based on various sources

Given these circumstances, the report “STI Outlook 2023” published by the OECD in March 2023 points out that “securitization” of STI policies is progressing. The term “securitization” here refers to the recognition of policy issues (climate change, food, energy, large-scale disasters, mineral resources, water, inter-technologies, emerging technologies, etc.) that were not fully considered as security issues in the past. The report also emphasizes the concepts of “Technology Sovereignty” and “Strategic Autonomy” as STI policy frameworks in major countries and regions (e.g., China, the EU, and the US). The report points out that three types of policy intervention tend to be implemented in science and technology⁵⁶.

- Protection: Restricting technology flows and reducing dependency risks through export controls and other regulatory policies, supply chain diversification measures, etc.
- Promotion: improving domestic innovation capacity and performance through comprehensive innovation policies, mission-oriented innovation policies, national industrial strategies, etc.
- Projection: Expanding and deepening international STI collaboration through international technology partnerships, active participation in international standardization bodies, etc⁵⁷.

Furthermore, the report states that these changes are creating a new environment for STI policy and emphasizes the need to transform the goals and instruments of STI policy. In particular, the report recommends the following points as important to strengthen the resilience function of STI and clarify the direction of STI policy and international cooperation⁵⁸.

- Governments need to treat issues cross-ministerially and coordinate on an issue-by-issue basis.
- One-size-fits-all measures should be avoided and strategic competition should be examined and addressed on a case-by-case basis.
- There is a need for “strategic intelligence” such as horizon scanning, foresight, and technology foresight assessments to make well-informed decisions in a turbulent and uncertain

56 OECD, “Chapter 2 Science, technology and innovation policy in times of strategic competition,” Science, Technology and Innovation Outlook 2023, (pp.43-86)
<https://www.oecd.org/sti/science-technology-innovation-outlook/>

57 The “European Economic Security Strategy” of the European Union also has a similar concept, but uses the term “partnering” instead of “projection”.

58 OECD, “Science, Technology and Innovation Outlook 2023,”
<https://www.oecd.org/sti/science-technology-innovation-outlook/>

environment.

- Like-minded governments should work together to avoid situations such as an inefficient “subsidy race” competition.

3.2.2 Overview of Efforts to Ensure Research Security in Major Countries

As discussed in the previous section, while strategic international collaboration on science and technology has become increasingly important, the damage to the health of the research system and its negative impact on national security due to technology outflows and other consequences of the undue use of open research systems has become apparent since around 2018. This has affected cross-border scientific collaboration and the international flow of scientists, which until then had been taken for granted for the exploration of new knowledge. Maintaining the right balance between scientific openness and scientific protection is now a key global issue in the context of science and technology policy⁵⁹. As of February 2025, there are more than 200 research security-related initiatives from 27 countries listed in the OECD policy database⁶⁰, compared to 17 from 12 countries in 2018. Below is an overview of the governments, policies, and initiatives of the United States, United Kingdom, Canada, France, and Germany, based on their public documents.

United States

Government Policy:

In accordance with the 1985 National Security Decision Directive (NSDD-189)⁶¹, all results of fundamental research conducted by government agencies, including the Department of Defense, are available to the public. In January 2021, President Trump (during his first administration) issued National Security Presidential Memorandum-33 (NSPM-33)⁶². This required funding agencies to establish a “Research Security Program” for research institutions that receive federal research funding. About three and a half years after the Biden administration took over, the Office of Science and Technology Policy (OSTP) issued guidelines in July 2024. In August 2022, the “CHIPS and Science Act”⁶³ was enacted to further strengthen research security as well as to protect critical domestic technologies such as semiconductors and promote research and development of critical technologies. As of February 2025, following the inauguration of the second Trump administration in January 2025, there have been no new developments, but future changes in policy should be closely monitored.

Dialogue between government and academia:

In 2020, the National Academies established the “Science, Technology, and Security

⁵⁹ Mervis, Jeff. “SCIENCEINSIDER.” Science, February 16, 2024.
<https://www.science.org/content/article/why-white-house-taking-so-long-issue-new-research-security-rules>. など

⁶⁰ OECD, “EC-OECD STIP Compass.” <https://stip.oecd.org/stip/research-security-portal>

⁶¹ <https://irp.fas.org/offdocs/nsdd/nsdd-189.htm>

⁶² <https://trumpwhitehouse.archives.gov/presidential-actions/presidential-memorandum-united-statesgovernment-supported-research-development-national-security-policy/>

⁶³ <https://www.congress.gov/bill/117th-congress/house-bill/4346>

Roundtable”⁶⁴ under the National Defense Authorization Act to promote dialogue between key personnel from federal agencies, intelligence agencies, and national security agencies involved in science and technology, and universities, government research institutions, and industry.

Academia in Motion:

In 2019, the independent scientific advisory group JASON released a report on “Fundamental Research Security”⁶⁵. The report notes that international collaboration is essential to the development of science in the United States, but that some foreign governments are using unfair means while enjoying the value of research collaboration. On the other hand, excessive regulation of basic research may hinder the development of science, advocating the need for a common understanding between government and academia. In 2024, the National Academies issued a report, “International Human Resource Development Program in a Changing Global Environment”⁶⁶. The report stated that there are risks involved in accepting foreign nationals, but that risks should be mitigated and international collaboration with like-minded countries and the Global South should be promoted, and recommended 11 items to attract talent to national security and defense-related fields.

Specific measures:

Funding Agencies: In June 2024, the National Science Foundation (NSF) announced a new research security initiative (TRUST) in the funding adoption process involving primary assessment of sensitivity through keyword searches, in-depth investigation of possible high risk, and implementation of risk mitigation measures as necessary. The trial was scheduled to begin in the quantum field in FY2025. OSTP issued guidelines in July 2024, and research institutions (including universities) that receive \$50 million or more in annual federal science and technology support are required to establish a research security program based on the OSTP guidelines. Such programs must include training in (1) cyber security, (2) international travel security, (3) research security, and (4) export control.

In support of ensuring research security, in July 2024, NSF announced the establishment of the “Safeguarding the Entire Community of the U.S. Research Ecosystem” (SECURE) under the CHIPS and Science Act with a five-year, \$67 million (approximately 10 billion yen)⁶⁷ budget. The research community-led initiative will share and provide solutions (such as shared tools, best practices, training, analysis, and other information) to support decision-making in the context of malignant foreign threats.

United Kingdom

Government Policy:

In 2019, the National Protective Security Authority (NPSA, formerly CPNI) released the

⁶⁴ <https://www.nationalacademies.org/our-work/national-science-technology-and-security-roundtable>

⁶⁵ https://www.nsf.gov/news/special_reports/jasonsecurity/JSR-19-21FundamentalResearchSecurity_12062019FINAL.pdf

⁶⁶ The National Academies of Sciences, Engineering, and Medicine (The National Academies)
<https://nap.nationalacademies.org/catalog/27787/international-talent-programs-in-the-changing-globalenvironment>

⁶⁷ <https://new.nsf.gov/news/nsf-backed-secure-center-will-support-research>

“Trusted Research Guidance”⁶⁸ on information disclosure and risk assessment of disclosed information. It is expected that individual universities will use this as a guideline and take measures in consideration of their own environment and other factors. In 2021, the National Security and Investment Act established an investment review system for 17 areas of specific sensitive technologies⁶⁹, requiring more stringent due diligence for the 17 areas. In response, the Cabinet Office issued its “Guidance for Higher Education Institutions and Research Institutes”. (The 17 areas to be reviewed are: advanced materials, advanced robotics, AI, nuclear power, telecommunications, computer hardware, major government supply, cryptographic authentication, data infrastructure, national defense, energy, military and general-purpose products, quantum technology, satellite and space technology, emergency services supply, synthetic biology, and transportation)

Dialogue between government and academia:

In April 2024, then Deputy Prime Minister Oliver Dowden received Vice Chancellors from a group of 24 UK research universities (the Russell Group) and was briefed on security by the Director General of MI5 and the Chief Executive of the National Cyber Security Center (NCSC)⁷⁰. He warned that sovereign states are targeting UK universities to achieve their own priorities, putting national security at risk. With regard to cutting-edge technology, the risks are particularly serious, he said. At the same time, Deputy Prime Minister Dowden also promised to maintain the openness and independence of the academic sector.

Academia in Motion:

In July 2024, the Russell Group released a position paper⁷¹ stating that UK universities have responded to threats by working closely with government ministries and security agencies, strengthening internal defenses with enhanced due diligence, staff training, and strong policies in line with guidance, but noted that a budget is needed to building on these responses and responding quickly to new threats.

Specific measures:

Funding Agencies: In 2021, UK Research and Innovation (UKRI) issued guidance⁷² stating that universities and research institutions receiving research funding are expected to have a system of due diligence in place during international collaborations. Based on this guidance, the EPSRC under the UKRI (which funds engineering research) has decided to conduct due diligence on research participants when supporting international collaborative research, with the aim of reducing research security risks as much as possible while not hindering opportunities for international research collaboration.

Support to ensure research security: In May 2021, the current Department for Science, Innovation and Technology (DSIT) established the Research Collaboration Advisory Team

⁶⁸ <https://www.npsa.gov.uk/trusted-research-academia>

⁶⁹ <https://www.gov.uk/government/collections/national-security-and-investment-act>

⁷⁰ “Government to launch new consultation to protect UK universities from security threats,” 26 April 2024. <https://www.gov.uk/government/news/government-to-launch-new-consultation-to-protect-uk-universitiesfrom-security-threats>

⁷¹ <https://russellgroup.ac.uk/policy/policy-documents/briefing-research-security-a-shared-responsibility/>

⁷² <https://www.ukri.org/manage-your-award/good-research-resource-hub/trusted-research-and-innovation/>

(RCAT), an organization that advises universities and research institutions on international security risks related to international research collaboration.

Canada

Government Policy:

The “National Security Guidelines for International Research Cooperation”⁷³ (2021) was released, outlining the nature of the risks and the direction of risk mitigation measures. The Guidelines provide a wealth of information on potential risks in international research collaboration, including a list of (1) sensitive research areas and (2) research partners of concern (designated research institutions), and a portal site “Safeguarding Your Research”⁷⁴ has been launched. In January 2024, the Government of Canada issued a “Policy on Sensitive Technology Research and Concerned Affiliations,” which clearly states that researchers who are affiliated with or have a background with institutions that may pose a risk to national security, or who receive funding or in-kind support from such institutions, may not receive research grants from the Canadian government to promote highly sensitive technology research areas.

Dialogue between government and academia:

In 2018, a working group was established at a gathering of government agencies, funding agencies, and university officials⁷⁵ to begin discussions on research security and promote related policies with the understanding of academia. In addition, the Science and Technology Advisor himself has sent out messages to raise the importance of research security.

Specific measures:

Funding Agencies: As a pilot phase of the implementation of the “National Security Guidelines for International Research Cooperation” (2021), risk assessment was piloted in 2021-2023 in certain open call programs of the Natural Sciences and Engineering Research Council (NSERC). When working with the private sector, applicants submit a risk assessment form, and the National Security Division provides additional assessment and advice as needed as a result of the risk assessment. If risks are identified, risk mitigation measures are developed. Science merit and risk assessment reviews are conducted separately⁷⁶.

Federal support for ensuring research security: The Research Support Fund (approximately \$25 million per year, established by the Social Sciences and Humanities Research Council (SSHRC), the Natural Sciences and Engineering Research Council (NSERC), and the Canadian Institutes of Higher Education and others that receive research funding from the Canadian Institutes of Health Research (CIHR)) supports efforts necessary to ensure security by providing a variety of tools, including building research security training courses, establishing a research

73 <https://science.gc.ca/site/science/en/safeguarding-your-research/guidelines-and-tools-implement-researchsecurity/national-security-guidelines-research-partnerships#2>

74 <https://science.gc.ca/site/science/en/safeguarding-your-research/guidelines-and-tools-implement-researchsecurity/sensitive-technology-research-and-affiliations-concern/policy-sensitive-technology-research-andaffiliations-concern>

75 <https://science.gc.ca/site/science/en/safeguarding-your-research/general-information-research-security/aboutgovernment-canada-universities-working-group>

76 <https://www.rsf-fsr.gc.ca/home-accueil-eng.aspx>

security center as a consultation center, presenting case studies, and introducing dual-use technology trends by the Ministry of Defense⁷⁷.

France

Government Policy:

In 2012, the “Potential Protection Scientific and Technical” (PPST) system was put into operation with the aim of safeguarding technical protection from a security and economic perspective. Among the targeted technological fields*, the PPST protects technology, knowledge, and information that may lead to (1) economic benefits, (2) military weapons, (3) proliferation of weapons of mass destruction, and (4) terrorism, and is positioned to complement the export control and foreign investment control systems already in place. Penalties under the Penal Code apply for violations. *Fields covered: biology, medicine, health sciences, chemistry, mathematics, physics, agriculture and ecology, earth, space, and space science, information and communication technology, and engineering. Institutions that deal with protected technology, knowledge, and information must establish restricted areas (ZRR), and physical and non-physical access is permitted only within those areas. Institutions (including universities) with restricted areas (ZRR) are assigned Security civil servants (FSD), who assist in the identification and assessment of risks at the institution. The FSDs are networked and work with the Security and Defense Officials (HFDS) located in each ministry and agency to implement the PPST.

Specific measures:

Ministry of Higher Education: A department has been set up to conduct risk assessments of research laboratories at research institutions, identify laboratories with high sensitivity (designated as restricted areas (ZRR)), and check whether the management of participants and information access is adequate.

The French National Research Agency (ANR): For the past two years, the ANR has been checking whether the candidate proposals for adoption include (1) overseas offices or (2) companies. If applicable, data is sent to the Ministry of Higher Education for a decision.

The National Centre for Scientific Research (CNRS): Ethics Officer (from the National Defense) has been appointed to serve as the point of contact with the Ministry. Each region and university also has personnel in charge of research security. Laboratories designated as confidential by the Ministry (including participants, management of research themes, etc.) are protected in cooperation with the Ministry.

Support for ensuring research security: The security officer (FSD) of each institution/university functions as a point of contact and a system is in place to consult with the responsible department of the Ministry of Higher Education.

Germany

Government Policy:

⁷⁷ <https://science.gc.ca/site/science/en/safeguarding-your-research/guidelines-and-tools-implement-researchsecurity/risk-assessment-review-process>

In 2023, the federal government developed its first security strategy⁷⁸ and China strategy. China has developed greatly economically and in terms of science and technology, and is too big to ignore, and while it is unrealistic to completely cut off collaboration, the key question is how to mitigate risk and collaborate with China. In March 2024, the Federal Ministry of Education and Research (BMBF) released a position paper on research security, recommending a policy of cooperation between BMBF and academia, security authorities, federal states, etc. under the principle of “as open as possible, as closed as necessary,” as well as a review of the adequacy of the civilian clause of universities (internal regulations that prohibit military research).

Academia in Motion:

In 2014, the German Association for the Promotion of Research (DFG) and Leopoldina (German Academy of Sciences) compiled guidelines for the handling of dual-use research, which were revised to include research security in 2022. The guideline states, “In principle, the researcher conducting the research and the research institution to which the researcher belongs should be aware of the risks and take responsibility for promoting the research.”

Specific measures:

Funding Agencies: In September 2023, the German Association for the Promotion of Research (DFG) developed and published “Recommendations for Addressing Risks in International Collaborations,” a set of guidelines to raise awareness among researchers about the benefits and risks of implementing international collaboration. In January 2024, the German Academic Exchange Service published its “Guidelines for Cooperation with China”. The guidelines state that cooperation with China itself, with its strongly developed scientific system, is important, and recommended three basic guidelines and specific recommendations for action: (1) be interest-oriented, (2) consider risks, and (3) cooperate in a manner appropriate to their capabilities (including the expansion of knowledge). It also established Kiwi, an organization to advise German universities on how to proceed with international cooperation (including risks).

3.2.3 Expected Role of Science and Technology Diplomacy

Approaches to ensuring research security differ with each country’s STI policy, systems, and historical background. However, in the international framework of like-minded countries, shared values and principles are being promoted to ensure that such differences do not impede international collaboration. The G7 Science and Technology Ministers’ Meeting (Bologna, July 2024) stressed the “growing importance of promoting research security integrity in international cooperation” and noted that “the G7 is an important forum for discussing, promoting and disseminating common values and principles and best practices”. At the OECD-CSTP (Committee on Science and Technology Policy) ministerial meeting (Paris, April 2024), the ministerial declaration included strengthening shared values in international cooperation, such as research security and integrity. In May 2024, the EU Council of Ministers in charge of research adopted a “Recommendation on Strengthening Research Security” for Member States. The

⁷⁸ <https://www.nationalesicherheitsstrategie.de/%E5%9B%BD%E5%AE%B6%E5%AE%89%E5%85%A8%E4%BF%9D%E9%9A%9C%E6%88%A6%E7%95%A5-JP.pdf>

recommendations include consideration of academic freedom, adherence to the principle of “as open as possible, as closed as necessary” in cooperation with non-EU member states, non-discrimination and respect for fundamental rights, strengthening cross-sectoral cooperation within governments, and establishing new support organizations.

In a new report published by The Royal Society and AAAS in February 2025, “Science diplomacy in an era of disruption” (see Chapter 1, 1.2), research security is identified as a new and challenging area of science and technology diplomacy. The report states, “While it is not yet clear how research security efforts will affect the process and outcomes of science diplomacy, which must balance the value of protection and international cooperation for national security purposes, research in this area is making progress.” Coordinating national rules for research security so as not to impede international collaboration in science and technology will be an important new role for science and technology diplomacy⁷⁹. In addition, although discussions on research security are currently focused on advanced Western countries, it is necessary to expand the discussion to the Global South in the future, and science and technology diplomacy is expected to play a role in this regard as well.

⁷⁹ Kana Asano and Tateo Arimoto, “Science Diplomacy — 15 Years On Special Issue,” October 17, 2024.
<https://www.sciencediplomacy.org/perspective/2024/research-security-for-actionable-science-diplomacy>

4 | Science and Technology Diplomacy in the International Framework

4.1 International Network for Governmental Science Advice (INGSA)

The International Network for Governmental Science Advice (INGSA) was established in 2014 at the International Council of Scientific Unions (ICSU, now the International Council for Science (ICS)) in Auckland, New Zealand, in 2014. It is a global network of policy makers, practitioners, academics, researchers, and others participating on an individual basis to develop and share theoretical research and practical methods and processes on scientific advice and bridging science, policy, and society. In recent years, the Network has been building capacity in the science-policy interface in developing countries and accumulating a database of collected and analyzed examples of scientific advice in various countries during the COVID-19 pandemic. To further this, they are actively organizing international symposiums, workshops, etc⁸⁰. INGSA was reorganized in 2021 from an affiliate of ICSU to an NGO corporation under New Zealand law, with the aim of achieving a more stable organization and operation. Currently, the President is from Canada and the Vice-Presidents from the United Kingdom, Ethiopia, and Chile. Membership is growing and currently stands at approximately 3,000 members⁸¹. Regional chapters have been established in Asia (Malaysia), Africa, and Latin America, and expansion to North America, Europe, and the Middle East is being considered. INGSA, in collaboration with ISC, FMSTAN, and others, is actively promoting the development of various methods of scientific advice, sharing of knowledge and experience, and training of young people in the relationship between science, politics, and society, both for common global benefit and for individual regions and national cultures, in order to respond to the new era of SDGs and pandemics.

INGSA has held an international conference every 2-3 years since the first one in Auckland. (Figure 7).

Figure 7 INGSA International Conference Achievements

Year held	Venue	Main theme
2014	Auckland	Science Advice to Governments: A global conference for leading practitioners
2016	Brussels	Science and Policy-Making: Towards a New Dialogue
2018	Tokyo	Science Advice in a changing world

⁸⁰ INGSA <https://ingsa.org/>
⁸¹ INGSA “INGSA Strategic Plan 2022-2025,” <https://ingsa.org/wp-content/uploads/2023/08/INGSA-Strategic-Plan-2022-2025.pdf>

2021	Montreal	Build Back Wiser : Knowledge, Policy, and Publics in Dialogue
2024	Kigali	THE TRANSFORMATION IMPERATIVE

Prepared by CRDS based on various sources

In May 2024, the 5th INGSA International Conference was held in Kigali, Rwanda under the main theme “The Transformation Imperative: Expanded Evidence for Inclusive Policies in Diverse Contexts”. For the first time since the COVID-19 pandemic, the event was held in a face-to-face format⁸². This was the first time the event was held in the Global South, and according to the organizers, 6,000 people (including online) from 65 countries attended. Throughout, equity, diversity, and inclusion in scientific advice were emphasized. President Rémi Quirion (Scientific Advisor to the Province of Quebec, Canada) emphasized the importance of scientific advice and science diplomacy to governments as we approach the 10th anniversary of the establishment of INGSA, and the importance of the participation of various sectors, not only at the national level but also at the international, regional, and city levels, as well as human resource development. Kristiann Allen, INGSA General Secretary, introduced the diverse activities of INGSA’s regional chapters. The Prime Minister of Rwanda and the Minister of Research and Education also gave speeches, stating that Rwanda aims to be a developed country by 2050, that it places great importance on investing in science, technology, and STEM education, that scientific advice is an important initiative for Rwanda, and that they welcome the INGSA conference. A keynote session and about 15 breakout sessions were held. The breakout session titled “Challenges of Science Diplomacy in a Divided World” featured Motoko Kotani, Deputy Chief Science and Technology Advisor to the Japanese Foreign Minister, Patricia Gruber, then Science and Technology Advisor to the US Secretary of State, and Marco Müller, Coordinator for Science Diplomacy and Multilateral Relations, European Commission, DG Research and Innovation. Officials from India, South Africa, China, and Brazil took to the stage and confirmed the importance of international collaboration using science and technology to solve global issues. In the session on “Institutional Development/Mapping - From Mechanisms to Ecosystems”, Dr. Yasushi Sato (Professor at Niigata University, CRDS Fellow) from Japan shared a case study on the SPI (Science Policy Interface). Tateo Arimoto (Counselor, JST; Visiting Professor, National Graduate Institute for Policy Studies) is a board member, and Kana Asano (JST and CRDS Fellow) served as a program committee member for the 2024 INGSA meeting.

At the November 2024 INGSA Board meeting, it was decided that the next INGSA meeting will be held in Malaysia in 2027.

82 INGSA, “INGSA2024: The Transformation Imperative Expanded Evidence for Inclusive Policies in Diverse Contexts,” <https://ingsa.org/ingsa2024/>

Figure 8 Concepts and Topics of the 5th INGSA International Conference

Topic	Topic Overview
Transformation	How should the principles and practices of scientific advice change as problem framing and scientific advice authority moves beyond technical diagnosis and solutions to individual policy problems to broader, more complex transformative approaches.
Expanded evidence	Scientific advice must help decision makers analyze and manage the complex issues facing societies across borders, and this requires expanding and applying evidence beyond traditional definitions and boundaries.
Inclusion	In order to (re)build and maintain trust in institutions and the social cohesion they engender, diverse audiences must be involved in the processes that affect them.

Prepared by CRDS based on various sources

4.2 Foreign Ministries Science & Technology Advice Network (FMSTAN)

The Foreign Ministries Science and Technology Advice Network (FMSTAN) is a worldwide informal network of science and technology advisors appointed by the foreign ministers of their respective countries⁸³. It was launched following an exchange of views on science and technology diplomacy held at the National Academy of Sciences in 2016, presided over by the Science and Technology Advisor to the U.S. Secretary of State. The four founding members are Vaughan Turekian, former Science and Technology Advisor to the U.S. Secretary of State; Peter Gluckman, former Chief Science Advisor to the Prime Minister of New Zealand; Teruo Kishi, former Science and Technology Advisor to the Ministry of Foreign Affairs; and Robin Grimes, former Chief Science Advisor to the British Foreign Office. FMSTAN is comprised of member countries’ principal science and technology advisors and science and technology advisors to their ministers of foreign affairs, as well as senior science and technology officials in their respective ministries of foreign affairs. In Japan, the Science and Technology Advisory System to the Minister of Foreign Affairs and the Council for the Promotion of Science and Technology Diplomacy were established in 2015. Currently Yoichiro Matsumoto serves as the Science and Technology Advisor to the Minister of Foreign Affairs and Motoko Kotani is the Deputy Chief Science and Technology Advisor to the Minister of Foreign Affairs. In November 2023, Patricia Gruber, Science Advisor to the U.S. Secretary of State, assumed the FMSTAN chairmanship, and as of February 2025, with her resignation of the position of Science Advisor to the Secretary of State, Allison Schwier, Acting Science Advisor to the U.S. Secretary of State, is serving as chair of FMSTAN. In light of the increasing global importance of science and technology in diplomacy, information sharing and policy dialogue have focused on the following three areas.

- (1) Sharing the science and technology diplomacy activities of each country

83 International Network for Government Science Advice, “Foreign Ministries Science & Technology Advice Network (FMSTAN),” <https://ingsa.org/divisions/fmstan/>

- (2) Sharing best practices and lessons learned in strengthening the science and technology advisory function
- (3) Strengthening the science and technology advisory function in ministries of foreign affairs and raising awareness of the importance of science and technology throughout the ministries.

The FMSTAN meetings held to date are shown in Figure 9.

Figure 9 FMSTAN face-to-face meetings in recent years (information exchange also conducted online on an irregular basis)

Date held	Venue
September 2017	Talloires, France (Tufts University European Center)
May 2018	Geneva, Switzerland (held consecutively with the United Nations Conference on Science, Technology and Development (CSTD/UNCTAD))
November 2018	Tokyo, Japan (held in conjunction with the International Network for Governmental Science Advice (INGSA) Conference)
February 2019	Muscat, Oman
November 2019	Vienna, Austria (in succession with WSF in Budapest)
March 2022	Cadarache, France
October 2022	Geneva, Switzerland (held in conjunction with the Annual Meeting of the Geneva Foundation for Scientific Diplomacy and Foresight (GESDA))
May 2024	Kigali, Rwanda

Prepared by CRDS based on various sources

4.3 American Association for the Advancement of Science (AAAS)

The American Association for the Advancement of Science (AAAS), founded in 1848, is a private, non-profit organization and the largest academic organization in the world. It was born out of an attempt to organize scientists throughout the country at a time when the number of scientists in the United States was still small. It is said to be the first organization in the world to promote science and engineering. The purpose of its activities is to provide useful science-related information to science, technology, and education professionals and various professional organizations. AAAS headquarters are located in Washington, D.C., with a staff of approximately 200. The association has approximately 1.2 million members, and membership is open to anyone who pays a membership fee.

In addition to publishing the journal *Science*, one of the two major scientific journals, AAAS holds an annual meeting around February each year. The meeting features scientific sessions, workshops, and poster sessions on the latest scientific findings, as well as a diverse program of symposia, where scientists and the general public discuss science, technology, and policy. The

meeting also features events and professional development sessions to promote connections among the diverse communities of scientists, educators, policymakers, and journalists. The 2025 Annual Meeting was held February 13-15 in Boston under the theme “Science Shaping Tomorrow”. In the chaos immediately following the inauguration of the second Trump administration, AAAS executives stressed the importance of strengthening the scientific enterprise from a long-term perspective in an era when the objectivity and truth of science are under threat. They strongly insisted that the scientific community must take proactive action in light of the current fragile state of the U.S. scientific ecosystem⁸⁴.

In 2008, AAAS established the AAAS Center for Science Diplomacy to build bridges between the international community, local communities, and societies and nations through close interaction between science and diplomacy. The Center regularly holds trainings and workshops on science diplomacy and offers an introductory online course. In addition, the Center promotes the practice of science advice internationally, presenting the “AAAS Science Diplomacy Award” to science or foreign policy communities for their contribution to the role of science cooperation. Since January 2024, in collaboration with The Royal Society, the Center has been leading an effort to rethink the framework for science and technology diplomacy, and played a central role in the publication of a new report, “Science diplomacy in an era of disruption,” in February 2025. (The report is described in detail in Chapter 1, 1.1).

The “Science and Technology Policy Fellowship” program implemented by AAAS has a 50-year history. The program sends scientists and engineers (many of whom hold doctoral degrees) to one of the legislative, judicial, or administrative branches of government for one year in principle, to gain hands-on experience in the process of science and technology policy making. Currently, about 300 scientists and engineers are dispatched each year to federal agencies, Congress, and the offices of legislators for about a one-year term. The number of applicants usually exceeds 800 and the hiring rate is about 30%. Salaries and activity fees are covered by the host institution when the fellowship is to a federal agency, and by AAAS or an academic society (partner association) when the fellowship is to a congressional office. Many of the approximately 4,000 graduates are active in the field of science and technology policy. (After completing their fellowships, about 42% remain with federal agencies, about 24% return to academia, and the remaining 34% work for non-profit or for-profit organizations.) The fellowships have also influenced the formation of similar programs in the U.S. and abroad; for example, the Association of Southeast Asian Nations (ASEAN), in partnership with the U.S. Agency for International Development (USAID), has offered the AAAS Science and Technology Policy Fellowship since 2014⁸⁵.

⁸⁴ AAAS, “AAAS Leaders: We Must Stay Focused Amid Turmoil” <https://www.aaas.org/news/aaas-leaders-we-must-stay-focused-amid-turmoil>

⁸⁵ AAAS, “Fellowship,” <https://www.aaas.org/fellowships>

4.4 Science and Technology in Society forum

The Science and Technology in Society forum (STS Forum) is an international conference on science and technology organized by the STS Forum, a non-profit organization. It is considered to be the world's largest international conference of its kind, focusing on issues related to science, technology, and society that are common to the world. Former Minister of Finance and former Minister of Science and Technology Policy Koji Omi conceived the idea in 2002 to create a science and technology version of the Davos Forum, and the annual meeting has been held at the Kyoto International Conference Center since 2004. The STS Forum was established in 2006 as a non-profit organization. The STS Forum Annual Meeting takes place every year on the first Sunday through Tuesday of October and is attended by approximately 1,500 world-class scientists, policy makers, and business executives from 120 countries, regions, and international organizations⁸⁶.

The 21st Annual Meeting was held in October 2024, bringing together nearly 1,500 leaders from more than 80 countries, regions, and international organizations⁸⁷. The opening session, "The World in 2024 -- What do we need from S&T?" was chaired by STS Forum President Komiyama and featured a video message from Prime Minister Shigeru Ishiba. STS Forum President Komiyama emphasized the traditional STS Forum theme of "Lights and Shadows" of science and technology, with a particular emphasis on AI and explained that there would be four keynote sessions on AI, including themes such as "AI and Medicine," "AI and Education," and "Ethics and Regulation of AI," as well as the timely creation and reading of discussion summaries by AI, in collaboration with the University of Tokyo. The conference declared its commitment to providing a forum for fair and open discussion by making full use of cutting-edge AI technology. Noting that while generative AI is bringing revolutionary changes to society, risks of false information and privacy are emerging, Prime Minister Ishiba said Japan will lead the "Hiroshima AI Process" and share a vision of "safe, secure, and reliable AI" at the G7. He also noted that the increasing power consumption of data centers is a challenge and that Japan should work together with industry, government, and academia on production and research of advanced semiconductors and human resource development, as well as formulate national strategies for nuclear fusion and quantum technology to achieve a carbon-neutral society. Furthermore, with the growing importance of international science and technology cooperation and science and technology diplomacy, Ishiba advocated accelerating international human resource exchange (international brain circulation) while ensuring research security. He also took note of the Osaka-Kansai Expo to be held in 2025 under the theme of "Design Future Society for Our Lives". A breakout session on "Science Diplomacy" was held, chaired by Sudip Parikh, CEO of AAAS (American Association for the Advancement of Science), with Takao Someya (Professor,

⁸⁶ The Science and Technology in Society forum, <https://www.stsforum.org/>

⁸⁷ STS forum, "Executive Summary of the STS forum 2024 – 21th Annual Meeting," https://www.stsforum.org/file/2024/11/STS2024_Summary.pdf

University of Tokyo) as the Japanese speaker. After identifying factors hindering international cooperation in science and pointing out the importance of human resource development for the younger generation, discussion went on to the influence of politics on science and technology diplomacy, gaps in science data, and the importance of cooperation between diplomats and scientists. Also identified was the need to promote international mobility, education and financing, and collaboration with industry. Other sector-specific meetings, such as the Meeting of Heads of Research Institutions, the Meeting of Heads of Funding Institutions, and the Meeting of University Presidents, as well as various related meetings and top-level talks were held in conjunction with the STS Forum annual meeting, reaffirming the forum's importance as an effective platform for science and technology diplomacy. It is expected that various stakeholders will take advantage of this valuable opportunity in the future. In addition, the 14th Meeting of Heads of Funding Agencies in 2024 was held under the theme of "Promotion of Science and Technology Diplomacy"⁸⁸, and involved discussions of the roles and challenges of funding agencies.

4.5 World Scientific Congress(WSF)

In 1999, the United Nations Educational, Scientific, and Cultural Organization (UNESCO) and the International Council of Scientific Unions (ICSU, now ISC) jointly organized the World Conference on Science, which resulted in the "Universal Declaration on Science and the Use of Scientific Knowledge" (Budapest Declaration). Four visions were presented as the responsibilities of science and technology in the 21st century and their contract with society: "Science for Knowledge; Knowledge for Progress," "Science for Peace," "Science for Development," and "Science in Society and for Society". This was the outcome of a week of discussions among 2,000 representatives of the science and technology community from industry, academia, and civil society from around the world who gathered in Budapest, the capital of Hungary. Based on deep reflection on how in the 20th century, science and technology played a part in wars, caused pollution, and had a major negative impact on society, a shift was declared from the perspective of a monolithic view of the history of progress and methods, which assumes that society will naturally progress if "science for knowledge" is promoted. This was a major paradigm shift in the values, ideas, and methods of modern science and technology established over the centuries since the Age of Enlightenment.

The World Science Forum (WSF) has been held since 2003 under the joint sponsorship of the Hungarian Academy, UNESCO, ICSU, and the American Association for the Advancement of Science (AAAS) to discuss how to implement the Budapest Declaration and share concrete examples. The WSF brings together about 500 people from academic institutions and funding agencies, as well as researchers, and policy makers from around the world every two years to

⁸⁸ JST, "The Funding Agency Presidents' Meeting" <https://www.jst.go.jp/inter/others/symposium/fapm/fapm.html>

deepen discussions⁸⁹.

Since the 6th conference in 2013, the conference has been held alternately in Budapest and in developing countries in response to the rapid increase in global-scale issues such as climate change, biodiversity, and other sustainability-related issues, and the worsening impact on developing countries (Figure 10).

Figure 10 History of the “Budapest Declaration” and WSF

	Year held	Venue	Main theme
Predecessor	1999	Budapest (Hungary)	Declaration on Science and the Use of Scientific Knowledge (The so-called “Budapest Declaration.”)
1st	2003	Budapest (Hungary)	Knowledge and Society
2nd	2005	Budapest (Hungary)	Knowledge, Ethics and Responsibility
3rd	2007	Budapest (Hungary)	Investing in Knowledge: Investing in the Future
4th	2009	Budapest (Hungary)	Knowledge and Future
5th	2011	Budapest (Hungary)	The Changing Landscape of Science – Challenges and Opportunities
6th	2013	Rio de Janeiro (Brazil)	Science for Sustainable Global Development
7th	2015	Budapest (Hungary)	The Enabling Power of Science
8th	2017	Dead Sea (Jordan)	Science for Peace
9th	2019	Budapest (Hungary)	Science, Ethics and Responsibility
10th	2022	Cape Town (South Africa)	Science for Social Justice
11th	2024	Budapest (Hungary)	The science and policy interface at the time of global transformations

Prepared by CRDS based on various sources

The 11th Forum was held in Budapest, Hungary in November 2024 under the theme “The science and policy interface at the time of global transformations”⁹⁰. More than 1,100 participants from 122 countries discussed global trends shaping the future of science and the

89 WSF, “History of World Science Forum,” <https://worldscienceforum.org/contents/history-of-world-science-forum-110010>
90 Science Council of Japan, “World Science Forum 2024 1st steering Committee Meeting” <https://www.scj.go.jp/ja/int/haken/230911.html>

interaction between science, society, and policymaking, and at the end adopted the “Declaration of the 11th World Science Forum: The science and policy interface at a time of global transformations” (see Column 4). At the forum, in addition to the traditional discussions of trust in science and the policy interface of science, the keyword “science diplomacy” emerged and was included in the declaration. Although no session on science diplomacy per se was held at the plenary session, there were many discussions, especially in Europe, at the side events. During the closing session, it was announced that the next 12th Forum will be held in Indonesia.

Column 4

11th World Science Forum (2024) Declaration: The interface between science and policy in a time of global transformation (Summary)

** Declaration published on the website, summarized and excerpted by CRDS*

(Preamble Summary)

- The world faces a rapidly changing and uncertain future. Despite tremendous advances in science and technology, climate change and most of the Sustainable Development Goals are being inadequately addressed, conflict is increasing, and inequality is widening. There is an urgent need to increase scientific understanding and promote evidence-based, science-based, long-term thinking and decision-making.
- Efficient, reliable, and creative science-policy interfaces are especially important at this time of global transformation.
- Recognizes the evolving role of science diplomacy in linking science and policy to solve challenges and to promote science for peace.
- Recognizes the challenges of evidence-based policymaking, but notes the growing risks posed by disinformation and other practices that undermine important trust in science. Regrets that distrust and outright denial of science is embedded in some political and social agendas. Need to confront these risks and remove potential obstacles.
- Acknowledges the universality of science, but recognizes that its impact will be limited unless the benefits of its advances and applications are shared equitably.

(Declaration Excerpts)

- Calls upon governments at all stages of development to ensure that knowledge production and translation is supported as an important factor in the development and well-being of the nation. Academic and

scientific freedom must be realized and maintained, science must be kept as open as possible, and science and scientific researchers must be protected. We call on the scientific community to redouble its efforts to ensure the generation of reliable knowledge and practical solutions, and to eliminate harmful scientific practices in both knowledge generation and reporting.

- Urge the scientific community to better inform both civil society and decision makers about the power of science to disseminate information and its limitations. These activities should be recognized as an important aspect of the scientific community's mission.
- Should further integrate efforts to use and protect science within the multilateral system and through its components, working with the global scientific community through formal organizations, including those related to science diplomacy, among others.
- Urge governments, organizations, and companies to ensure that scientific developments and new and emerging technologies are subject to ethical standards.

4.6 Geneva Science and Diplomacy Anticipator(GESDA)

The Geneva Science and Diplomacy Anticipator (GESDA) is a Geneva-based foundation established in 2019 for science and technology diplomacy initiatives with the support of the Swiss government and others⁹¹. Its vision is to “Use the future to build the present” by bringing together different communities to jointly anticipate scientific and technological advances and develop comprehensive, global solutions for a sustainable future based on those advances. Initially set for a pilot period through 2023, it was decided in March 2022 to continue activities through 2033, with funding and other support from the Swiss government⁹².

The three main activities are: holding an annual meeting (Summit), identifying and evaluating scientific and technological advances expected in the next 5, 10, and 25 years through the preparation of the Science Breakthrough Radar report, and activities based on the report's suggestions. A distinctive feature of the Summit is the deep commitment of international organizations (the United Nations and related agencies). It was also announced that GESDA

⁹¹ GESDA <https://gesda.global/>

⁹² GESDA, “Science diplomacy: Federal Council continues to support GESDA Foundation after successful pilot phase,” <https://gesda.global/science-diplomacy-federal-council-continues-to-support-gesda-foundation-aftersuccessful-pilot-phase/>

plans to showcase future science and technology and its impact on society through an exhibition in the Swiss Pavilion at the 2025 World Expo in Osaka (Osaka-Kansai Expo)⁹³.

93 GESDA “Step into the Future with GESDA at the World Expo 2025, Osaka, Japan”
<https://gesda.global/step-into-the-future-with-gesda-at-the-world-expo-2025-osaka-japan/>

Conclusion ~Future Challenges and Prospects~

The Japan Science and Technology Agency (JST) established the Office for Research on International Trends in Science and Technology in September 2022 and the Science and Technology Diplomacy Group in the Center for Research and Development Strategy (CRDS) in April 2023 to strengthen the international strategic systems for STI policy. This report summarizes the latest global trends in STI policy from the perspective of science and technology diplomacy, based on the experience and analysis of this group in planning and participating in related international activities with the cooperation of relevant departments. The report concludes with some future issues and prospects.

Formation of the concept and framework of science and technology diplomacy

The interrelationship between science and technology and diplomacy has a tortuous history of light and shadow over the centuries as modern science and technology developed. In the 21st century this relationship has been conceptualized as “science and technology diplomacy”. An international workshop co-sponsored by The Royal Society and the American Association for the Advancement of Science (AAAS) in 2010 and attended by 15 countries agreed on a basic framework of objectives and methods for science and technology diplomacy. There are three types of diplomacy: diplomacy for science, science for diplomacy, and science in diplomacy. Since then, based on this, there have been diverse activities of science and technology diplomacy, from a strategic layer involving the formation of international networks of heads of state, ministerial-level science and technology advisors, and academia in each country, to a practical layer of concrete joint activities among the UN SDGs, research funding agencies, universities, and individual researchers.

Need for change in the framework of science and technology diplomacy

However, after 2020, the political, economic, technological, security, and geopolitical global environment have been drastically changing due to the COVID-19 pandemic, Russia’s invasion of Ukraine, the intensifying struggle for hegemony between the United States and China, the development and competition of emerging technologies such as AI, and their increasing social impact. These factors have made it imperative to reexamine the 2010 framework for science and technology diplomacy. The objectives, scope, and people and organizations involved have been expanded and diversified, and are currently in the process of being redesigned. As already described in detail in the main text, some of the challenges and prospects for the future are as noted below.

1. Diversification and expansion of the objectives and scope of science and technology diplomacy

- Global-regional-national-local and outer-space perceived as spatial levels.
- International joint research in basic sciences.
- International joint projects in science and technology that address difficult global issues

such as climate change.

- Joint projects to achieve value such as enhancing competitiveness and security in each country's national interest or in a limited number of countries (e.g., like-minded countries) or regions.

Science and technology diplomacy is entering a difficult period in which it is necessary to be strategic and selective in response to the objectives and the values it aims to achieve.

2. Diversification and expansion of actors and organizations involved in science and technology diplomacy: competition and collaboration, increased presence of the private sector

The organizations, groups, and individuals listed below are in dynamic motion, competing and cooperating with each other according to their various objectives. We live in an era that demands constant monitoring of trends and active participation.

- Politics, government, funding agencies and organizations, universities, businesses, NGOs, international organizations, etc.
- G7, G20, OECD, UNESCO, International Science Council (ISC), World Science Forum (WSF), International Network for Governmental Science Advice (INGSA)

3. 2025 will be a landmark year in the history of science and technology diplomacy

The year 2025 will be an important year for Japan, as shown below, with a series of international conferences and reports to be held and published to discuss the challenges and prospects of science and technology diplomacy (Figure 11). It is necessary to demonstrate Japan's contribution and presence in these conferences and reports, and it is in the national interest to do so. The CRDS Science and Technology Diplomacy Group was deeply involved in the discussions preceding the February 2025 publication of the AAAS and The Royal Society of London report, "Science Diplomacy in an Era of Disruption" (detailed in Chapter 1 of this text). This report reorganized the three frameworks of 15 years ago into two frameworks (Science impacting diplomacy and Diplomacy impacting science) with more emphasis on practice. It is expected to be a useful practical guide for researchers, practitioners, entrepreneurs, and their groups who are bridging the two areas of science and national interest, which are inherently very distant from each other, through international activities. Furthermore, a major international conference on science and technology diplomacy is scheduled to be held in Asia from 2026 to 2027, and Japan is expected to make a positive contribution in the inner circle by setting the agenda. The World Science Forum (WSF) (Indonesia) and the International Science Council (ISC, Beijing) will be held in 2026, and the International Network for Governmental Science Advice (INGSA) will be held in Malaysia in 2027.

Figure 11 Discussion forum related to science and technology diplomacy after February 2025

Date	Contents
February 2025	AAAS Annual Meeting (Boston, USA) AAAS and The Royal Society Announce New Science and Technology Diplomacy Framework
March 2025	UNESCO Ministerial Scientific Diplomacy Meeting (Paris, France)
May 2025	UN STI for SDGs: STI forum (New York, USA)
May 2025	Global Head of Funding Agency (GRC) Annual Meeting (Riyadh, Saudi Arabia)
Spring-Summer 2025	G7 Summit and related meetings (Canada)
Summer - Fall 2025	OECD STI Outlook 2025 to be announced
October 2025	STS Forum and related meetings (Kyoto, Japan)
November 2025	Geneva Foundation for Scientific Diplomacy and Foresight (GESDA) Annual Meeting (Switzerland)
2025	G20 Summit (also to host 3rd Scientific Advisory Roundtable) (South Africa)
December 2025	2nd European Science Diplomacy Conference and European Science Diplomacy Framework to be announced (in Copenhagen, Denmark)
2026	12th World Science Forum (WSF) (Indonesia)
2026	International Science Council (ISC) General Assembly (Beijing, China)
2027	International Network for Governmental Science Advice (INGSA) International Conference Malaysia

Prepared by CRDS based on various sources and information from related parties.

Secure and develop human resources for science and technology diplomacy

The UN, ISC, INGSA, and the European Commission have advocated that the promotion of new human resources for scientists and engineers in the 21st century should emphasize the development of designers, scientific advisors, and communicators who synthesize knowledge and create value, in addition to traditional scientists who create knowledge and produce papers. This is directly applicable to the image of human resources who will be responsible for science and technology diplomacy in the new era.

As we enter an era of dynamic and diverse changes in the currents of science and technology diplomacy, people and organizations engaged in this field in the severe international arena are required to analyze and act while constantly considering the values and norms of behavior and the 5W1H (who, what, for what, when, where, and how).

Science and Technology Diplomacy and Japan's Challenges and Prospects

For Japan, it is essential to maintain and improve its level of science and technology, to be a leader in emerging technologies such as AI, semiconductors, quantum, and communication technologies, as well as in international cooperation and brain circulation. It is imperative that Japan work to enhance competitiveness, secure energy, food, and markets, and to strengthen supply chains, as well as to expand science and technology diplomacy activities and international networks at various levels and help formulate a strategic selective response for

each country and region. Under the current situation of severe global division, the Global North, Global South, and Asia have high expectations for Japan. These expectations will not last long if we do not make efforts now. Strategic efforts for science and technology diplomacy must be made in unison by the government, industry, academia, and citizens. Securing and fostering excellent human resources and establishing viable career paths are also major issues that need to be addressed.

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RESEARCH REPORT

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